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Exploring a user-centered approach for movement-based features in interaction design

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HIGHLIGHTS

- Designer-interpretable features enable non-technical practitioners to utilize motion analysis.
- Spatial visualizations make group detection accessible without requiring algorithmic expertise.
- Five themes reveal how accessible features transform installation design practices.
- Designers prioritize spatial patterns over biomechanical precision in public spaces.
- Evidence-based iteration replaces technology-constrained design in motion systems.

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ABSTRACT

Far beyond very fine-grained, accurate data, movement-based interaction design often benefits from working with higher-level features to create engaging interactions. Yet, movement-based frameworks most often focus on the detection strategy without providing clarity on how to use such data for the design of user interactions. This purely technical approach makes it challenging for practitioners to explore feature extraction technology as a design material. This paper presents an approach that considers designers' perspectives and attitudes about using movement-based features to propose a set of *designer-interpretable* descriptors and enhance their ability to use human motion in interaction design. In addition, a computational prototype is utilized to visually present the features and help designers better understand movement content. The investigation into the effects of *designer-interpretable* features on the concept ideation and design of motion-based interactions followed a user-centered approach, and its validity was established through a qualitative study conducted with interaction designers. Semi-structured interviews following a creative practice exercise were employed to evaluate how the computational prototype influenced interaction designers' processes, allowing them to reflect on their experience and compare it to previous approaches when designing movement-based interactions. The study determined that interaction designers were able to leverage features to identify detection capabilities and enrich the ideation process. Moreover, feature visualization provided further insights into the characteristics of movement, which helped practitioners understand the interaction opportunities that come with it.

1. Introduction

Movement-based interaction design for public spaces faces an accessibility challenge: while sophisticated computer vision technologies can track multiple people, analyze group dynamics, and extract complex movement patterns in real-time (Sundaresan and Chellappa, 2005; Joo et al., 2017), the features these systems provide are typically expressed in technical terms that do not translate naturally into design concepts accessible to non-technical practitioners (Escamilla et al., 2021). This gap between technological capability and design accessibility becomes

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pronounced when interaction designers attempt to create experiences for room-scale environments such as museums, galleries, and public installations (Rogers and Muller, 2006; Michelis and Müller, 2011), where the scale and diversity of visitor behaviors demand both technical sophistication and creative design insight (Caserman et al., 2020; Chapman and Hume, 2021; Cosentino and Giannakos, 2023).

The significance of this accessibility challenge has grown alongside the broader evolution of Human-Computer Interaction (HCI) toward more natural, social, and user-centered approaches that move beyond traditional desktop paradigms (Dourish, 2001; Glonek and Pietruszka, 2012; Churchill et al., 2013). Full-body interaction, which involves capturing and analyzing body movements, posture, and physical gestures to control digital content, has emerged as a particularly compelling approach for creating immersive experiences that leverage the entire body as an interface (England, 2011). These applications span diverse contexts from gaming and education to art exhibitions, demonstrating their potential for creating engaging user experiences (Pensyl et al., 2008; Godbehere and Goldberg, 2014; Casillas-Perez et al., 2016). The concepts of full-body interaction, multiple-people motion sensing, room-scale spaces, and computer vision merge to define the motion-based interaction design space that this paper explores.

Computational frameworks for movement analysis have emerged as tools that consider feature extraction a relevant aspect of human motion understanding (Bevilacqua et al., 2007; Alemi et al., 2014; Amin and Burke, 2018). However, these systems focus on technical capabilities and detection strategies without addressing how to translate movement data into design opportunities (Bogdan et al., 2020). This limitation becomes problematic given the diversity of practitioners within interaction design, where various profiles of designers approach movement-based projects with different levels of technical expertise—a distinction that is crucial and explicitly addressed in this paper since it represents an important factor influencing successful system implementation (Sundström et al., 2011; Sørum, 2017). While some designers possess technical backgrounds, many others lack knowledge in feature extraction or motion analysis, creating barriers for practitioners who approach design from pragmatic or creative perspectives rather than purely engineering viewpoints (Fallman, 2003), yet who possess essential skills for creating user experiences. Although the accessibility of computer vision technology continues to improve, offering new possibilities for exploring innovative interaction paradigms (Bellotti et al., 2002; Eriksson et al., 2007), the translation of these technical capabilities into designer-friendly tools remains underdeveloped.

To address this accessibility gap, we propose an approach that prioritizes designer interpretability in movement-based feature extraction and visualization for public exhibition contexts. Our previous study with interaction designers revealed that practitioners working on installations for museums, galleries, and public spaces need movement analysis tools that progress systematically from basic technical reliability to creative exploration opportunities (Escamilla et al., 2021). These designers employ data-driven, analytical approaches where the design process involves observing how general audiences move through instrumented spaces, analyzing those movement patterns, and creating responsive systems based on insights derived from visitor behavior. A key insight from our previous work was that features should not only provide meaningful information for non-technical practitioners but also help them identify a clear path toward designing for these contexts. To encapsulate these conditions, we use the term *designer-interpretable* to describe features specifically tailored to the needs of interaction designers working in public exhibition spaces.

This research makes three contributions to understanding how interaction designers work with movement-based features when creating experiences for room-scale public installations. First, we introduce *designer-interpretable* features as a design principle—movement analysis capabilities that align with designers' spatial reasoning about visitor behavior rather than requiring algorithmic expertise. Our framework advances prior work on proxemic interaction (Greenberg et al., 2011) and group detection (Setti et al., 2015) by translating technical capabilities into design-actionable features through three dimensions: detection matching design intentions, technical-to-creative progression appropriate for diverse practitioner backgrounds, and spatial abstraction supporting public space design. Second, we identify principles for achieving designer interpretability through a research probe implementing seven features. Our work demonstrates that spatial visualizations, and progressive complexity enable designers to understand sophisticated visitor behavior analysis without technical training in computer vision or machine learning. Third, our qualitative study reveals how *designer-interpretable* features transform creative practice. Five themes show that accessible movement analysis shifts designers from technology-constrained iteration to evidence-informed exploration, enabling new approaches to spatial design, social interaction orchestration, and natural interface development in public exhibition contexts.

The study follows a user-centered methodology that combines computational prototype development with qualitative evaluation through interaction designers' creative practice. We present related work, first examining interaction designers' varying technical engagement levels and existing approaches to movement feature extraction and visualization specifically relevant to public exhibition design, and then introducing the design framework that guides the alignment of motion-based feature extraction tools with design processes for these contexts. We then provide a description of our computational prototype that implements multi-view 3D pose estimation, feature extraction, and visualization capabilities. We validate our approach through a qualitative study involving interaction designers engaged in creative practice exercises specifically focused on public exhibition space design and semi-structured interviews, through which we analyze how visualization tools influence design processes and compare our approach to practitioners' previous experiences with movement-based interaction design. The paper concludes with discussions, limitations, and future work.

2. Related work

2.1. Perspectives on interaction designers' technical engagement

Understanding how different interaction designers approach technology is crucial for developing effective movement-based design tools. The field encompasses practitioners with varying technical expertise, creating an accessibility challenge for sophisticated motion analysis systems. Sundström et al. (2011) identify critical patterns in how designers engage with technology, revealing that many adopt a 'black-boxed' approach where technology's unique properties remain unexplored. While designers are expected to navigate technological complexity, their focus typically stays on user behavior rather than technological possibilities. This emphasis leads to a limited examination of how technology's distinct characteristics could open new design opportunities.

Fallman (2003) provides a framework for understanding these differences through three competing accounts of design practice. The conservative account views design as a scientific endeavor, drawing on rationalism and systems theory. Under this process-oriented account, structure is at the heart of practicing design, and the designer's view is that of an engineer. The pragmatic account emphasizes interpretation and meaning creation through iterative design processes. It focuses on the situatedness of the designer in the life world and brings to light the interweaving of roles, practices, and technologies involved in design. The romantic account emphasizes creativity over technical considerations. It nourishes the idea of 'creative geniuses,' a legacy of the Enlightenment. Designers are seen as creative individuals with unusual talents who often face opposition to defend their artistic freedom.

To understand the practical implications of these design perspectives, we use the participants from our previous study (Escamilla et al., 2021) as a case study, revealing distinct patterns in how practitioners with different technical backgrounds approach their work (see Table 1 for details). Non-technical interaction designers often align with pragmatic or romantic accounts, with those following the romantic perspective

Table 1
Details of focus group participants (Escamilla et al., 2021).

Area of expertise	View on design (competing account)	Role model	Technical background	Gender
Audiovisual communication	Romantic	Artist	Non-skilled	Female
New media artist	Romantic	Artist	Non-skilled	Female
Digital arts research	Romantic	Artist	Non-skilled	Male
Creative coder	Conservative	Engineer	Strong	Male
Creative director	Conservative	Engineer	Strong	Male
Interaction developer	Conservative	Engineer	Intermediate	Male
VR content creator	Pragmatic	Designer	Non-skilled	Male
Creative director	Pragmatic	Designer	Intermediate	Male
Cognitive systems and interactive media	Conservative	Engineer	Strong	Male
User experience design	Pragmatic	Designer	Non-skilled	Male
Full-body interactive experience design	Pragmatic	Designer	Non-skilled	Female
Cognitive systems and interactive media	Pragmatic	Designer	Non-skilled	Male

typically working in audiovisual communication, new media art, and digital arts research, focusing on crafting visually engaging and artistically meaningful user experiences. Those following the pragmatic approach—VR content creators, creative directors, and full-body interactive experience designers—concentrate on ensuring their designs resonate effectively in real-world scenarios while spanning interactive media direction, and user experience design for embodied interactions. In contrast, designers with strong technical backgrounds who align with the conservative view typically work as creative coders, developers, or in cognitive systems, possessing the programming expertise to directly engage with algorithmic concepts.

This diversity in designer profiles creates a fundamental accessibility challenge in movement-based interaction design, where the complexity of computer vision systems and feature extraction algorithms can overwhelm practitioners who lack technical backgrounds, preventing them from leveraging powerful analytical capabilities in their creative practice. While technically-oriented designers can utilize existing frameworks, the broader community of interaction designers remains unable to fully harness sophisticated movement analysis technologies, highlighting the critical need for approaches that prioritize designer interpretability alongside technical capability (Sundström et al., 2011).

It is important to clarify the specific design practice this research addresses within the broader landscape of movement-based interaction design. Our work focuses on supporting designers who employ data-driven, analytical approaches to creating movement-based experiences for public installations such as museums, galleries, and event spaces. In these contexts, designers create experiences where general audiences—individuals without specialized movement training or performance backgrounds—interact through their movements in instrumented spaces. The design process typically involves capturing and analyzing how visitors naturally move through and occupy spaces, identifying patterns in their spatial behaviors and social formations, and iteratively developing responsive systems informed by these empirical insights. This analytical design approach complements other legitimate methodologies in the movement-based interaction design landscape, such as bodystorming where designers physically enact movements to explore interaction possibilities (Márquez Segura et al., 2016a), somaesthetic design workshops that prioritize first-person embodied experience (Hook, 2018), and choreographic methods that draw on dance and performance expertise (Fdili Alaoui et al., 2015). While these approaches emphasize experiential and performative dimensions of movement, our work addresses the complementary need for tools that enable designers to systematically analyze and understand movement patterns in their target environments, particularly when designing for diverse public audiences whose movement behaviors cannot be predetermined through designer enactment alone. This focus shapes both the features we develop—emphasizing spatial relationships, group dynamics, and traffic patterns over biomechanical precision or expressive qualities—and our evaluation methodology, which examines how movement analysis tools influence design thinking for public exhibition contexts.

2.2. Movement feature extraction

For designers creating movement-responsive installations, understanding the landscape of available feature extraction approaches is essential for making informed tool selection decisions. The landscape of movement feature extraction spans diverse technical approaches and application contexts. Computer vision advances have enabled sophisticated multi-person tracking (Shuai et al., 2022; Kong et al., 2022), with comprehensive surveys (Dang et al., 2020) documenting the breadth of available sensing modalities. While these systems demonstrate robust technical capabilities, they typically output low-level data that requires substantial expertise to interpret for interaction design purposes. Motion analysis research has similarly developed powerful methods for temporal pattern recognition (Patrona et al., 2018), yet these operate at granularities that challenge non-technical interpretation.

Recognizing the need for more accessible approaches, several toolkits have emerged to provide higher-level abstractions of movement data for creative practitioners. Camurri et al. (2000) pioneered features targeting expressive movement qualities, extracting characteristics like Fluency, Directness, Contraction Index, and Quantity of Motion to capture movement's qualitative dimensions. Building on this foundation, Mova (Alemi et al., 2014) offers an interactive movement analytics platform that combines skeletal pose descriptors with machine learning capabilities, enabling designers to train custom movement classifiers. However, its reliance on joint-level features and machine learning concepts may pose interpretability challenges for practitioners unfamiliar with biomechanical analysis or algorithmic training. OpenMoves (Amin and Burke, 2018) provides insights into movement patterns through clustering and person tracking, implementing algorithms for hotspot detection and group formation analysis. While offering valuable analytical capabilities, effective use depends on familiarity with machine learning concepts and algorithmic parameter tuning. Modosc (Dahl and Visi, 2018) presents real-time movement descriptors based on marker positions, calculating features like velocity, acceleration, and bounding volumes that offer concrete insights into movement dynamics. Though more tangible than abstract learned features, translating these point-based descriptors into broader behavioral understanding and interaction design strategies requires additional interpretive work. InteractML (Hilton et al., 2021) addresses non-technical users through a node-based interface for designing movement interactions in Unity, democratizing access to interactive machine learning. Despite its accessible visual programming approach, the system still relies on underlying machine learning concepts that can create barriers to complete designer interpretability.

The spatial computing perspective provides complementary insights that have yet to be fully integrated with feature extraction approaches. Greenberg et al. (2011) demonstrated that proxemic interactions—organized around distance, orientation, movement, identity, and location—enable spatial computing that aligns with natural human behavior. Their proxemic media spaces and surfaces showed how focusing on spatial relationships rather than low-level motion data can create compelling interactions. However, their framework was implemented for specific applications rather than as generalizable, designer-accessible features. Similarly, Setti et al. (2015) provide rigorous sociological grounding for group detection through F-formations, showing that computational approaches informed by social science significantly outperform purely geometric methods. Yet the technical sophistication of their algorithms (graph-cuts, dominant sets) requires expertise that limits accessibility. Wu et al. (2017) demonstrated an intermediate approach with EagleSense, achieving 90.55% activity recognition through

simplified top-view sensing and extremity-based features rather than full skeletal reconstruction, though even this simplified system presents a 72-dimensional feature space requiring technical knowledge to utilize effectively.

The common challenge across these approaches is translating technical capability into practical design tools. While toolkits like Mova, OpenMoves, and InteractML make movement analysis more accessible than working directly with computer vision libraries, they still require an understanding of machine learning concepts, algorithmic parameters, or joint-level biomechanics. Spatial interaction frameworks (Greenberg et al., 2011; Setti et al., 2015) offer theoretically rich foundations but lack implementation as generalizable features. Even accessibility-focused systems (Wu et al., 2017) produce outputs requiring technical interpretation. This persistent gap between capability and usability—despite advances in tracking accuracy, feature sophistication, and spatial interaction theory—motivates the need for features explicitly designed for designer interpretability. Our work addresses this gap specifically for designers analyzing visitor behaviors in room-scale environments, where understanding spatial patterns, social dynamics, and traffic flows informs responsive system design.

2.3. Movement and feature visualization

Visualization plays a critical role in making movement data interpretable, with research exploring diverse approaches to representing motion capture data's multivariate, time-dependent nature (Alemi et al., 2014; Bernard et al., 2017; Liu et al., 2020; Arpatzoglou et al., 2021). For designers working with visitor movement data in installation contexts, visualization serves dual purposes: enabling analytical understanding of behavioral patterns and supporting creative exploration of interaction possibilities. These visualization challenges parallel broader information visualization principles (Agrawala et al., 2011; Tufte, 1985) while requiring specialized techniques for temporal and spatial data representation.

Two distinct visualization philosophies have emerged in movement-based interaction research. The first emphasizes objective, data-driven representation. Bernard et al. (2017) developed MotionExplorer, which uses multiple coordinated views to support detailed motion data analysis, enabling users to explore spatial, temporal, and relational aspects simultaneously through linked visualizations. Liu et al. (2020) demonstrated that motion graphs can effectively organize large datasets by representing movements as nodes with transitions between them, facilitating pattern discovery and sequence analysis. Arpatzoglou et al. (2021) extended these approaches with DanceMoves, combining traditional motion capture visualizations with novel analytical views specifically designed for dance movement analysis. These objective approaches prioritize analytical clarity and data fidelity, supporting systematic exploration of movement patterns.

The second philosophy emphasizes expressive, metaphorical visualization that captures movement qualities. In the context of using movement qualities in digital arts (Mentis and Johansson, 2013; Silang Maranan et al., 2014; Fdili Alaoui et al., 2017), conventional visualization techniques such as levels and numerical values often fall short, mainly due to their limited ability to capture the nuanced dynamics of performed movements (Alaoui et al., 2015). Consequently, there is an inclination toward employing more sophisticated and abstract visualization approaches that convey the intricacies of movement (Subyen et al., 2011; Alaoui et al., 2015). Although the qualities of movement inspired by Laban Movement Analysis (Camurri et al., 2005; Subyen et al., 2011; Silang Maranan et al., 2014) primarily focus on the expressive and enjoyable aspects of movement, it is essential to differentiate this focus from our research interest, which is primarily centered on analytic visualizations. This body of work delves into the creative and artistic dimensions of movement, seeking to enhance the experience of movement interaction, often with a focus on providing a relatively free-form and enjoyable experience. This resonates closely with domains such as dance, where movement interaction has been harnessed to elevate the experiences of professional dancers, promoting expressiveness and enjoyment.

Rather than the artistic aspects of movement, our research is inherently driven by a distinct purpose—developing visualizations tailored to the requirements of interaction designers working with general audiences. These designers typically target diverse populations who may lack extensive training in movement performance and the subtleties of movement qualities. Consequently, our visualizations are designed to cater to this broad audience context, aiming to facilitate designers' comprehension and analysis of visitor movement data. We acknowledge that these users may not possess the specialized knowledge or expertise in movement dynamics that trained performers bring. Therefore, our work aligns with a mission—providing analytical tools that empower interaction designers to derive valuable insights from movement data coming from diverse visitor populations in installation environments.

2.4. A design theory for designer-interpretable features

The gaps identified in existing literature—between technical frameworks and design accessibility, diverse designer backgrounds and tool requirements, and visualization approaches and design practice—necessitate a systematic framework to guide the development of movement-based features for interaction designers. Building on our previous research with practitioners creating experiences for room-scale environments (Escamilla et al., 2021)—which specifically examined designers working on public event installations—we present a design framework that addresses these identified limitations by establishing principles for creating *designer-interpretable* features. This framework articulates empirically-grounded design heuristics that emerged from iteratively analyzing practitioners' expressed needs and creative processes, distilling recurring patterns into actionable principles for developing movement-based features that serve designers with varying technical backgrounds.

Our framework development followed an iterative, data-driven approach grounded in designers' actual practice within public exhibition contexts. We began with thematic analysis of focus group data, then refined and validated these themes through follow-up discussions with participants from our original study, testing whether the framework captured designers' experiences, clarified their design challenges, and offered practical utility. As we developed and tested movement-based feature extraction tools, we continued refining our understanding through cycles of prototyping and evaluation, ensuring the framework remained grounded in actual design practice for room-scale environments.

The resulting design framework organizes designer needs along three critical dimensions that movement-based features must address when analyzing visitor behaviors in installation contexts. First, *detection capabilities* must align with what designers want to identify in user behavior. For single users, designers need to analyze movement in relation to space, discriminate between actions, and estimate gestures. For multiple users, designers require tools to identify relationships among users, detect people flow and grouping patterns, and analyze collective movement behaviors. Second, features must support a *technical-to-creative progression* (see Fig. 1) that begins with purely technical necessities—simplifying technology use, eliminating noise from non-user detections, and reducing user frustration—and advances toward creative exploration opportunities that foster innovation and open new design possibilities. Third, *accessible presentation* is essential given designers' varying levels of technical expertise. Features should be abstracted into higher-level concepts that align with designers' mental models rather than presenting raw motion data, with gradual introduction that matches typical design workflows, progressing from basic spatial features to temporal, qualitative, and finally combinatorial features.

This design framework directly informed the development of our computational prototype, with seven movement-based features designed to address specific aspects of these three dimensions. Section 3 presents our implementation, where the design rationale for each

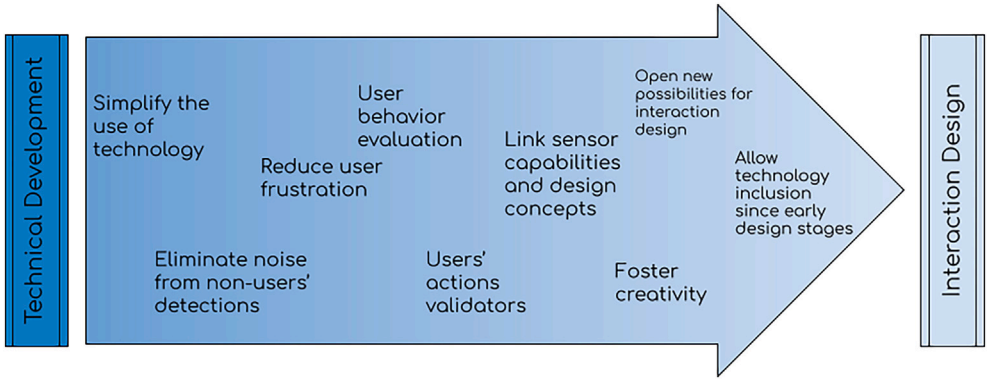


Fig. 1. Visual summary of interaction designer needs when using motion-based feature extraction tools.

feature—grounded in this framework—is presented alongside its technical specification and visualization approach, demonstrating how empirical designer needs translated into concrete system design decisions.

3. Implementation of a computational prototype

To investigate how *designer-interpretable* features might influence interaction design practice, we developed a computational prototype that serves as a research probe. The prototype's primary purpose is methodological: it provides a concrete, functional artifact through which designers can experience and reflect upon movement-based features in practice, enabling us to study how accessible motion analysis tools shape creative processes and design thinking.

Our prototype implements seven movement-based features that progress from individual spatial analysis to group behavior detection, each designed to embody the principles of our design framework. The features support understanding how people navigate spaces, where they congregate, how they form social groups, and what patterns emerge across multiple visitors—core concerns for designers creating responsive installations. These seven features specifically address installation design for general public audiences rather than movement qualities requiring specialized interpretation. This scoping distinguishes our work from movement analysis serving other legitimate purposes: ergonomic assessment integrating environmental context, expressive performance systems prioritizing Laban-like movement qualities, or biomechanical analysis demanding joint-level kinematic precision. Our contribution is the systematic investigation of how these capabilities, when presented through *designer-interpretable* abstractions and visualizations, affect the ideation and design of movement-based interactions for installation contexts. By implementing familiar motion analysis techniques, we create conditions for observing how designers with varying technical backgrounds engage with movement data, what insights they derive from feature visualizations, and how these tools influence their creative practice.

The prototype was developed specifically to support qualitative research activities—guided explorations, creative practice exercises, and reflective interviews—rather than for deployment in production design workflows. Designers interact with pre-recorded movement data through an interface that prioritizes interpretability over real-time performance, allowing them to explore feature behaviors, compare visualizations, and develop interaction concepts appropriate for installations serving general audiences. This research-oriented design enables us to isolate and study the relationship between feature presentation and design thinking, providing empirical evidence about which aspects of movement analysis prove valuable for interaction design practice and why.

What follows is an examination of the prototype's three primary components: a 3D pose estimation and tracking component, a feature extraction system, and the visualization engine. Complete hardware and software specifications for system replication are provided as Supplementary materials.

3.1. Pose estimation and tracking in 3D space

The 3D pose estimation and tracking component addresses the challenge of analyzing multiple visitors simultaneously in room-scale environments, employing a multi-camera approach to account for large volume spaces and resolve occlusions and limited field-of-view problems common in installation settings. The procedure for this component consists of a sequence of steps. First, a calibration step based on using a *ChArUco* board pattern to capture properly synchronized images, followed by obtaining camera poses through bundle adjustment optimization. Second, people detection and pose estimation on each view. To this end, we adopted the body-only *AlphaPose* estimator (Fang et al., 2022) which follows a top-down strategy. *AlphaPose* is a two-step framework that first detects human bounding boxes and then independently estimates the pose within each box. In the current implementation, an off-the-shelf *YOLOV3* pre-trained detector (Redmon and Farhadi, 2018) and an efficient high-accuracy pose estimator (Fang et al., 2022) are adopted. Third, a multi-view correspondence and 3D pose reconstruction step. To match the estimated 2D poses across views, we adopted the approach of Dong et al. (2019) in which appearance similarity and geometric compatibility cues are combined to calculate the affinity score between bounding boxes. The bounding boxes with no matches in other views are regarded as false detections and discarded. The multi-way matching algorithm groups 2D poses of the same person in different views from which 3D poses can be reconstructed (Belagiannis et al., 2016). Fourth, a tracking-by-detection and filtering step to finally obtain smoothed human poses of all the people in the scene. To improve the identification ability, we compute the appearance feature similarity between non-redundant candidates and existing tracks to address the spatial distance limitation. Temporal averaging is used to fill in missing joints, while a Gaussian kernel with standard deviation σ is used to smooth each joint trajectory (Tanke and Gall, 2019). Fig. 2 shows the output of the multi-view 3D pose estimation and tracking system where 15 body landmark locations are detected for each analysis prove valuable for interacti person.

3.2. Movement-based features

The prototype implements seven features that directly address the detection needs identified in our design framework. These features progress from individual movement analysis to group behavior detection, with each feature designed to provide both analytical capability for understanding visitor patterns and design exploration opportunities for creating responsive experiences.

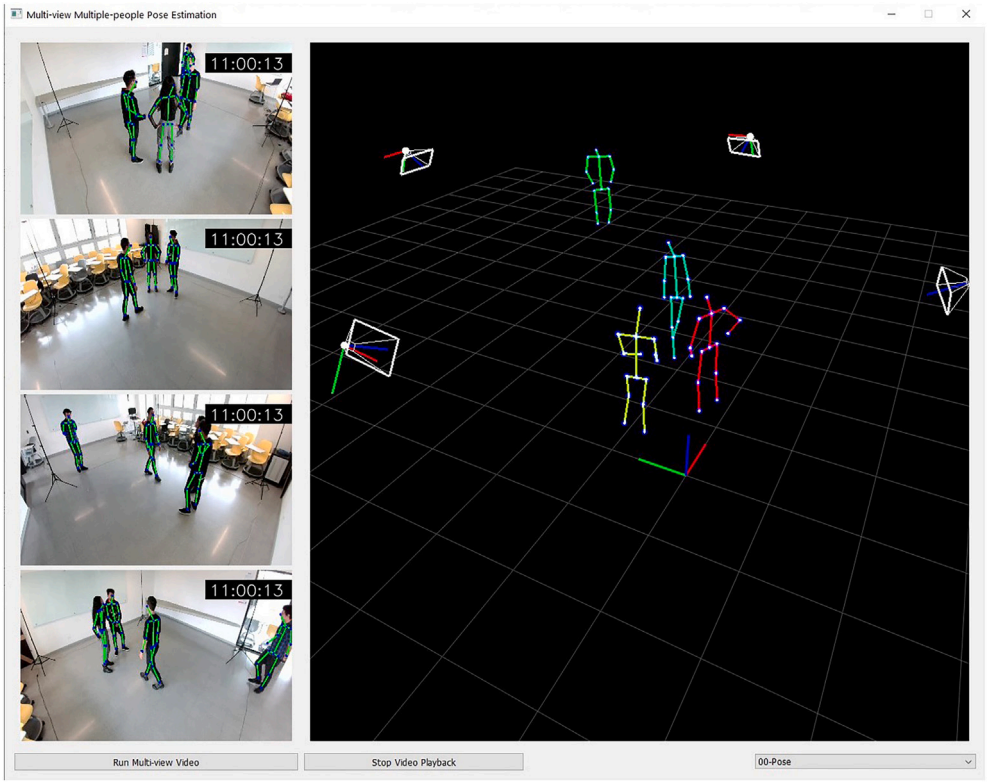


Fig. 2. Multi-view multiple-people tracking system.

3.2.1. Bounding box

Our design framework identified that designers need intuitive ways to analyze how users occupy and move through space—a fundamental detection capability for single-user spatial relationships. The bounding box feature directly addresses this need by providing immediate spatial understanding without requiring technical expertise in computer vision. This feature exemplifies the technical-to-creative progression principle by simplifying technology use through automatic calculation while enabling creative applications through its variation over time. The straightforward geometric representation supports accessible presentation, making spatial occupancy immediately interpretable regardless of technical background.

The bounding box feature calculates the rectangular parallelepiped enclosing the 3D pose of each person in real-time. While computationally straightforward, this representation supports rich spatial understanding without requiring technical expertise. The bounding box's variation over time enables designers to abstract useful representations of user motion, supporting both analytical and creative applications. This dual utility is evident in prior work, such as Glowinski et al.'s (2011) successful use of bounding shapes in analyzing affective body gestures. Fig. 3(a) shows how our visualization overlays the bounding box on a rendered 3D view of the scene, providing immediate visual feedback that designers can readily interpret and incorporate into their design thinking.

3.2.2. Trajectory

Responding to designers' needs to understand movement patterns and spatial pathways—a core detection capability for analyzing single-user movement in relation to space—the trajectory feature presents user movement as accessible temporal data. This feature supports the technical-to-creative progression by providing reliable movement history (a technical necessity) while enabling pattern recognition and design exploration (a creative opportunity). The short-time window approach exemplifies accessible presentation by focusing on manageable, recent movement rather than overwhelming designers with complete historical data, allowing them to grasp movement dynamics intuitively.

The trajectory feature presents user movement as a short-term characteristic that captures recent positional history. We implement this as a circular buffer of three-dimensional points, using the central point between shoulder joints as a stable reference for overall body movement. This choice reflects our understanding that designers need reliable reference points for movement analysis while maintaining flexibility in their interpretation. The trajectory data also serve as input for more complex group behavior analysis, demonstrating how our features build upon each other to support increasingly sophisticated analysis. Fig. 3(b) illustrates how our visualization makes these movement patterns immediately apparent while maintaining the spatial context designers need for their work.

3.2.3. Heading

Our design framework revealed designers' need to understand users' directional intent and orientation in space—essential for discriminating between user actions and predicting movement. The heading feature addresses this detection capability by approximating directional intent in an immediately interpretable form. This feature demonstrates technical-to-creative progression by reducing complex velocity calculations to simple directional indicators (simplifying technology) while enabling anticipatory interaction design (opening new possibilities). The visualization through fixed-size arrows exemplifies accessible presentation, providing intuitive directional information that aligns with designers' natural understanding of movement and space.

The heading feature addresses this by approximating the linear velocity vector of the user trajectory using the difference between the last two position points. We deliberately focus on the ground plane projection, discarding the z-dimension, as our research shows designers primarily think about directional intent in terms of horizontal movement. This simplification aligns with our understanding that features should reduce complexity while maintaining meaningful information.

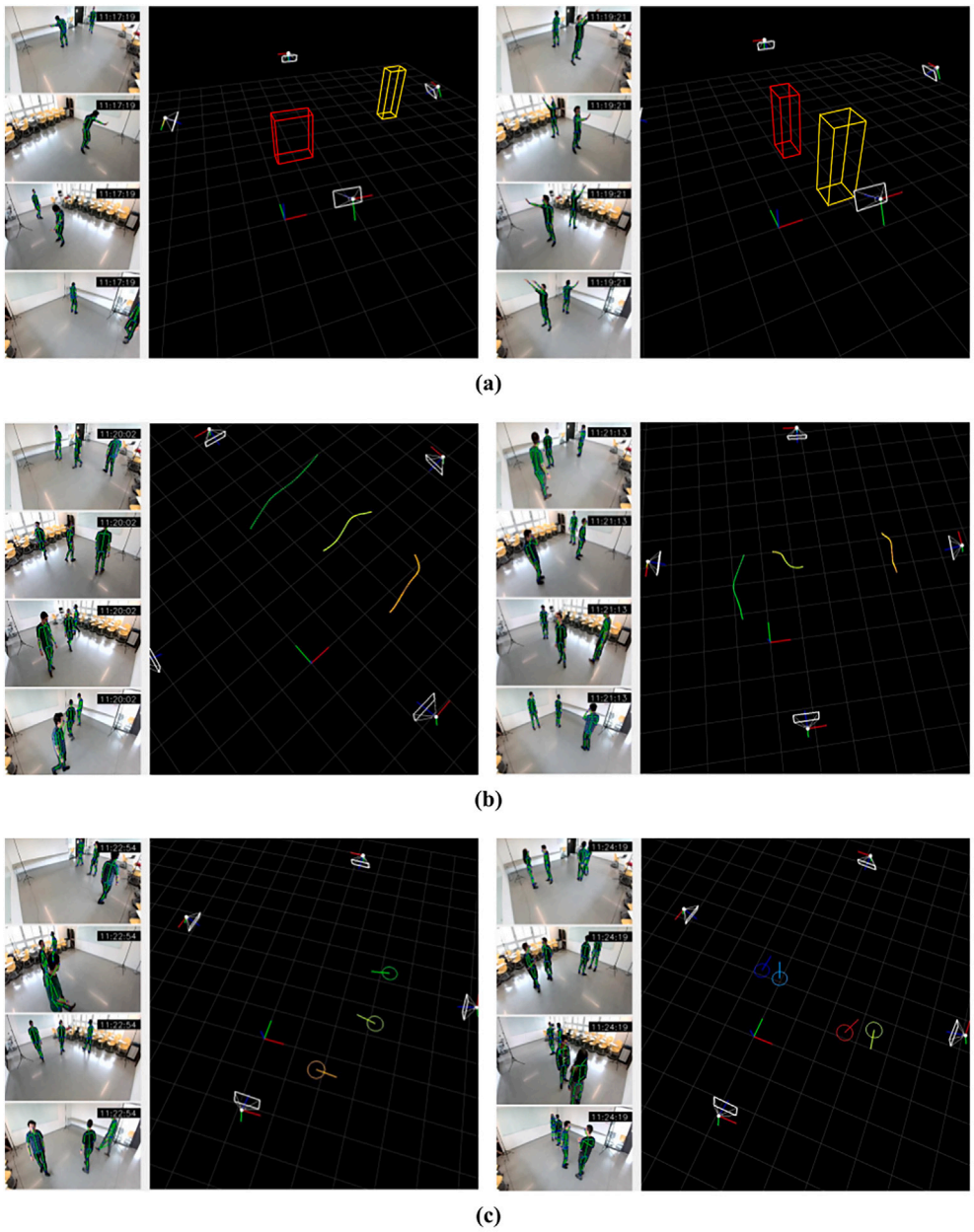


Fig. 3. Visualizations of individual movement features. (a) bounding box, (b) trajectory, and (c) heading.

By assuming forward movement and expressing heading as an angle between the velocity vector and the x-axis, we provide an intuitive representation that designers can easily incorporate into their interaction models. The visualization presents this information through fixed-size arrows emanating from current user positions, as shown in Fig. 3(c), offering immediate visual feedback about movement direction.

3.2.4. Instantaneous clustering

Moving from individual to group analysis, our design framework highlighted designers' critical need to understand real-time group formation and social dynamics—a key detection capability for identifying relationships among users. The instantaneous clustering feature employs unsupervised learning to address this need, directly supporting the technical-to-creative progression by eliminating the technical necessity of pre-configured parameters while fostering creativity through adaptive social pattern recognition. The dual visualization strategy—connecting lines for membership and circles for scale—exemplifies accessible presentation by making abstract clustering algorithms immediately comprehensible through familiar geometric representations.

The instantaneous clustering feature employs the mean shift algorithm (Comaniciu and Meer, 2002) to identify natural groupings based on users' centroid positions at each time step. The mean shift algorithm's ability to discover clusters without predefined numbers aligns with our understanding that social interactions in public spaces are fluid and unpredictable. This approach builds upon previous work by Amin and Burke (2018), who demonstrated its effectiveness in analyzing social interaction scenes and live performances. Our visualization strategy represents cluster membership through ground plane lines connecting users to their cluster centroids, while a circle radius indicates group sizes. This dual representation, shown in Fig. 4(a), provides designers with an immediate understanding of both individual group relationships and overall social distribution.

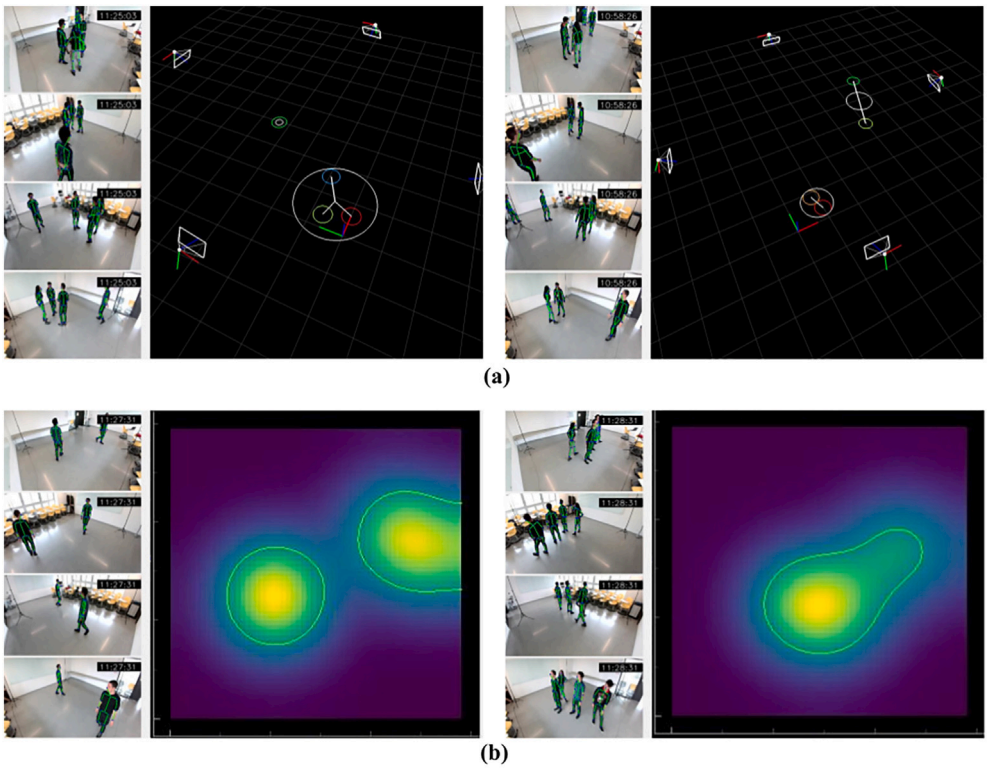


Fig. 4. Visualizations of user clustering features. (a) instantaneous clustering, and (b) hotspots.

3.2.5. Hotspots

Our design framework emphasizes designers' need to understand spatial usage patterns over time—a crucial detection capability for analyzing people flow and grouping patterns. The hotspots feature addresses this by identifying frequently visited areas and routes, providing macroscopic analysis that would be impractical through manual observation. This feature exemplifies the technical-to-creative progression by handling technical necessities like density-based clustering and outlier filtering automatically, while opening new creative possibilities through pattern revelation that enable evidence-based spatial design decisions. The heatmap visualization demonstrates accessible presentation by transforming complex temporal-spatial data into an intuitive visual form that communicates position, spread, and boundaries without requiring algorithmic understanding.

Based on the work of Amin and Burke (2018), hotspots are addressed as a long-term, macroscopic form of clustering. Nevertheless, we decided to perform clustering on positional data over a fixed time window using a different algorithm. We chose the DBSCAN algorithm (Ester et al., 1996) for its ability to perform density-based clustering, which is robust to outliers, aligning with our understanding that designers need reliable pattern detection even in noisy public environments. The algorithm's assumption that clusters are dense regions separated by areas of lower density proved particularly effective for hotspot visualization. Our visualization approach employs a heatmap representation, (see Fig. 4(b)), which enriches designers' perception of group behavior by making temporal patterns spatially evident, supporting both analytical understanding and creative exploration of space utilization.

3.2.6. Trajectory similarity

Building on designers' need to understand coordinated behaviors—a sophisticated detection capability for analyzing collective movement patterns—the trajectory similarity feature detects instantaneous common patterns across user trajectories. This feature supports the technical-to-creative progression by automatically handling complex mathematical analysis to reduce user frustration with technical complexity, while fostering creativity by revealing synchronization patterns that might inspire collaborative or competitive interaction designs. The event-based output format exemplifies accessible presentation by translating complex continuous similarity scores into clear binary detection events that designers can readily incorporate into interaction logic.

By performing Procrustes analysis (Goodall, 1991; Barhorst-Cates et al., 2022) between trajectory pairs, we create a similarity matrix that yields a mean score of movement coordination. The analysis is deliberately made scale, rotation, and reflection invariant, ensuring that similar movement patterns are recognized regardless of users' absolute positions. This implementation choice reflects our theoretical understanding that designers are often more interested in movement qualities and relationships than precise spatial matching. We format the feature as an event-based variable, providing clear positive detections when the mean score exceeds a threshold. As shown in Fig. 5(a), this binary output simplifies the complex mathematical analysis into actionable design information.

3.2.7. Correlations across movement patterns

This feature extends trajectory similarity analysis to address designers' needs to understand nuanced inter-user relationships and detect which specific users perform synchronized movements at any given moment. Rather than producing a single aggregate detection event, this granular approach directly supports sophisticated interaction design by revealing unconscious social coordination patterns between subsets of users. The feature represents the culmination of our framework's technical-to-creative progression, building upon simpler individual features to open new possibilities for complex social interaction design. The visualization graphically indicates corresponding users through visual indicators at their instantaneous positions, as shown in Fig. 5(b), exemplifying accessible presentation by making complex multi-user relationships immediately apparent through visual clarity rather than requiring technical expertise or algorithmic understanding. This feature provides

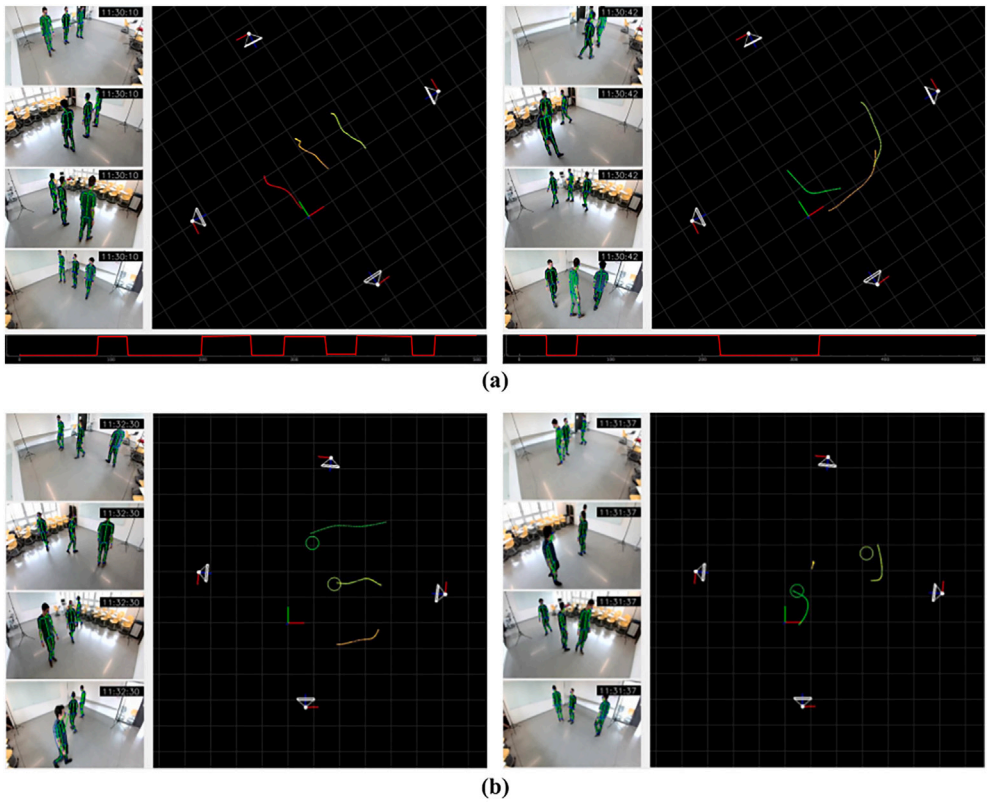


Fig. 5. Visualizations of group behavior features. (a) trajectory similarity feature, showing user trajectories and event-based detection at the bottom of the window, and (b) correlations across movement patterns, with circles indicating instantaneous position of users performing synchronized movements.

designers with tools to explore complex social coordination while maintaining the immediate utility and interpretability that characterize our entire feature set.

3.3. Visualization engine

The visualization engine embodies the conceptual understanding of how features should be presented to interaction designers, creating a visual environment that makes complex visitor movement data accessible regardless of technical background. The graphical user interface, shown in Fig. 6, consists of three sections that reflect our understanding of how designers engage with movement data. The *selection* section provides controls for opening scenes and choosing features, supporting the gradual introduction strategy. By allowing designers to isolate specific features, this approach prevents cognitive overload and enables focused exploration of movement characteristics. The *multi-view video playback* section overlays 2D pose detection results on each camera view, addressing designers' need to validate system behavior by directly comparing detected poses with actual movement. This visual validation builds trust in the system and provides immediate feedback about detection accuracy, supporting designers' evaluation needs.

The *three-dimensional representation* section forms the core of the visualization strategy, rendering feature visualizations within a spatial context that designers can freely navigate. This design decision directly responds to our finding that designers need to understand spatial relationships from multiple perspectives to fully grasp interaction opportunities when planning installation layouts. While we deliberately limited direct algorithm parameterization, we embraced Sundström et al.'s (2011) concept of "inspirational bits" by enabling experiential exploration through other means. The prototype allows designers to "feel" and experience the material qualities of movement data through spatial navigation, perspective shifts, and feature selection rather than numerical parameter adjustments. The implementation supports six degrees of freedom in camera movement through keyboard and mouse input, enabling designers to explore relationships between features and space from various viewpoints—zooming in to examine interactions or panning out to observe overall patterns. This spatial exploration becomes a form of material engagement, where designers develop a tacit understanding of movement data's properties through active investigation rather than abstract parameterization.

By shifting the mode of engagement from parameter tuning to spatial exploration and feature comparison, we create an environment that benefits practitioners with non-technical backgrounds, allowing them to engage with complex visitor movement data through familiar spatial and visual thinking rather than algorithmic concepts. Throughout the visualization engine design, we adhered to Tufte's conception of graphical integrity, ensuring that visual representations neither overrepresent nor underrepresent the underlying data (Tufte, 1985). This balance between technical accuracy and designer accessibility creates a visualization environment that supports both analytical understanding and creative exploration, fulfilling the dual needs identified in our working theory.

4. Design of the evaluation strategy

To evaluate how our computational prototype enhances the ability of interaction designers to use human motion in their practice, we conducted a qualitative study that combined usage evaluation with creative practice. Our primary objective was to investigate whether the proposed movement-based features might be helpful for interaction designers working on experiences for room-scale environments where diverse audiences interact through their movements, and how these features could be incorporated into their design work for such contexts. Since our study followed a user-centered approach, we employed a qualitative

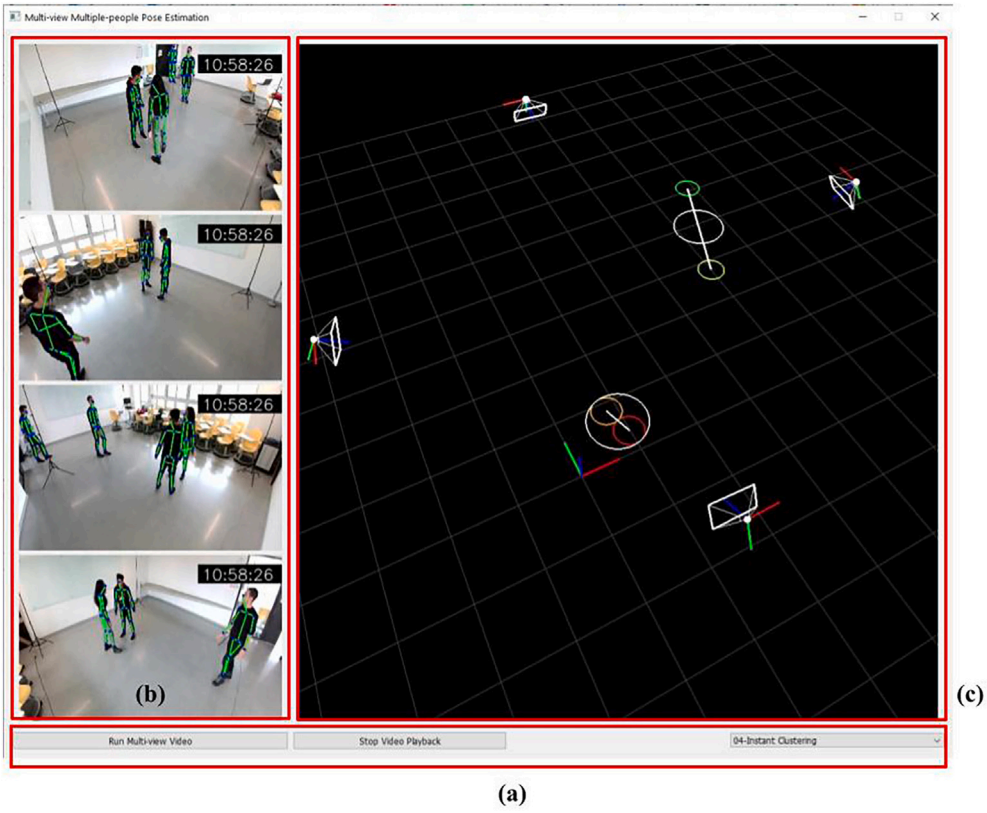


Fig. 6. Graphical user interface of the feature extraction and visualization prototype. (a) selection section, (b) multi-view video playback section, and (c) three-dimensional representation of the scene and feature visualization section.

methodology to develop an in-depth understanding of interaction designers' needs, perceptions, and creative processes when working with movement data.

4.1. Participants

We recruited five interaction designers with professional experience in designing movement-based interactions for installations and interactive experiences. This selected group represented diverse backgrounds within interaction design practice, including specialists in creating new interfaces, developing interactive systems, producing new media experiences, and contributing to interactive digital arts. All participants possessed prior expertise in developing interactive full-body experiences based on movement, which was essential for meaningfully evaluating our prototype's capabilities. While this sample size is modest, our decision to work with five participants was intentional, prioritizing depth over breadth in our qualitative investigation. This approach aligns with established practices in HCI research where rich, detailed insights from expert practitioners often prove more valuable than broad surveys (Ball and Ormerod, 2000; Cegarra and Hoc, 2006; Wirfs-Brock et al., 2021; Erete et al., 2023). However, we acknowledge that this limited sample size may restrict the generalizability of our findings to the broader interaction design community.

The participants represented a balanced distribution of technical orientations, with two having extensive programming experience and having previously developed interactive motion-tracking systems, while three identified primarily as designers with limited technical backgrounds who focused more on conceptual and experiential aspects of interaction design. The age range of the participants spanned from 24 to 45 years ($M = 31.8$, $SD = 9.9$), and their professional experience varied from 4 to 15 years ($M = 7.8$, $SD = 4.7$), ensuring a robust foundation of design knowledge and practical expertise in creating experiences for diverse audiences.

The evaluation was conducted through individual semi-structured interviews, which allowed for in-depth exploration of each participant's thought processes, creative approaches, and evaluative perspectives without the influence of group dynamics. This one-on-one interview format enabled participants to express their opinions freely without the conformity pressures or negotiated perspectives that often emerge in group settings. Through these focused individual sessions, we gathered particularly nuanced insights into each designer's experiences with the prototype, their design reasoning, and their critical assessment of the tool's capabilities. Though the sample size is modest, the richness and depth of data obtained through these semi-structured interviews provide substantial qualitative evidence regarding the prototype's impact on interaction design practice.

4.2. Study preparation

A room-scale interaction space was configured for the study, with a total area of approximately 5m x 4m. The space was equipped with four high-definition cameras placed at various positions, always above the height of people, to capture movements from different perspectives. This multi-camera configuration was designed to record each view and enable the tracking of multiple people in a space with occlusions and limited field-of-view problems. Before the participant sessions, we collected a comprehensive set of movement data from this environment, capturing natural interactions between individuals in various scenarios relevant to installation spaces where general audiences interact. This data included recordings of people exploring the space individually and in groups, with varying levels of engagement and interaction patterns representative of visitor behaviors in museums, galleries, and similar contexts. Each multi-view recording was then processed using the prototype's 3D pose estimation and tracking component, extracting 15 body landmark locations for each individual in 3D space. The processed data enabled the rendering of a 3D view of each scene with people's poses

accurately overlaid in the virtual space. This setup would serve as use cases during the study, allowing participants to interact with the feature extraction system during a guided walkthrough. Participants would be able to select specific movement features from a list and observe their visual rendering during playback, providing a hands-on understanding of how the system analyzes and represents human movement.

The study was structured in two main phases: a guided exploration and learning phase followed by a creative practice. During the guided exploration phase, sessions progressed through four steps. First, we presented an overview of the study structure and set expectations for the creative task. Second, participants received an introduction to the prototype, including its purpose and underlying technology. Third, we provided a short demonstration of the basic features, showing how the system processes and visualizes movement data. Fourth, participants engaged in a guided walkthrough of different features using the pre-recorded videos from natural interaction settings, with an incremental introduction to more complex features. As participants gained familiarity with the prototype, we progressed from simple individual movement visualizations to more complex group interaction features.

In the creative practice phase, participants worked on a realistic interaction design challenge representative of their professional practice. They were instructed to "Ideate an interactive experience for a public exhibition space that responds to the movement patterns experienced." This framing was deliberately chosen to align with the installation design contexts our participants typically work within while allowing creative freedom in their approach. During this phase, participants had access to the full pipeline of prototype usage, from recording new movement data to exploring the complete range of motion-based features and their visualizations. Participants viewed pre-recorded movement scenes and could record new movement data, observing both through the prototype's interface: synchronized multi-view camera feeds with pose detection overlays, a 3D visualization space showing skeletal representations and extracted features, and feature selection controls allowing them to isolate and compare individual movement characteristics. We provided materials for sketching and documenting their design concepts but imposed no constraints on the form or presentation of their ideas. This approach allowed us to observe how the prototype might integrate into their natural design processes and influence their creative decision-making.

4.3. Interview method and data collection

Following the creative practice phase, we conducted semi-structured interviews with each participant to gather insights about their experience with the prototype and analyze their design outputs. Each interview, lasting between 90 and 120 minutes, was audio-recorded for subsequent analysis. The interviews were structured to explore three primary areas: how the prototype influenced the participant's design process, what challenges and opportunities they identified while using the system, and how their experience compared to their previous approaches to designing movement-based interactions without such a tool.

The interview protocol was designed to encourage reflection on specific aspects of the prototype's impact on their design thinking. Participants were asked to walk through their design process step by step, highlighting moments where the prototype either enabled new insights or presented obstacles. They were prompted to explain their decision-making regarding which features they chose to incorporate into their designs and why. To facilitate this reflection, we reviewed with participants the design artifacts they created during the creative practice phase, asking them to elaborate on how specific movement features influenced their design decisions and concept development for the installation scenario. Representative examples of participant design concepts, and the complete interview protocol are available in the Supplementary materials.

Our data collection strategy followed two complementary approaches, both implemented through the semi-structured interviews. First, we conducted a comparative analysis of design outputs to evaluate three key dimensions: the sophistication of motion-based interactions proposed by participants, their integration of multiple motion features into cohesive interaction concepts, and the practicality and originality of their design solutions. Rather than merely examining the artifacts created, we used the interviews to explore participants' rationale behind their design choices and how the prototype's features contributed to their conceptual development. Second, we performed a usage pattern analysis to understand participants' engagement with the prototype itself. Our interview protocol included specific questions about which features participants found most valuable, how their usage patterns evolved throughout the design session, and what workflows emerged as they gained familiarity with the system. The reflective discussions during interviews provided rich insights into participants' reasoning behind their feature selection and usage patterns, allowing us to understand not just which features they used but also why they found them valuable in their design process.

By integrating these analytical approaches within the interview framework, we were able to develop an understanding of not only what the prototype enabled designers to create but also how it influenced their creative and technical processes. The semi-structured interview format allowed participants to articulate connections between the tool's capabilities and their design outcomes, providing rich qualitative data about the prototype's potential impact on interaction design practices for movement-based systems.

5. Analysis and results

The semi-structured interviews were analyzed using reflexive thematic analysis considering its systematic yet flexible approach to identifying patterns of meaning across qualitative data (Braun and Clarke, 2021). Coding was conducted by the corresponding author as the primary analyst, with themes developed through iterative reflection rather than seeking consensus between multiple coders. This approach aligns with reflexive thematic analysis methodology, which emphasizes the researcher's active role in theme construction and does not require inter-rater reliability measures, as the analytical process is inherently interpretive rather than seeking objective truth (Braun and Clarke, 2021). A total of 556 minutes were collected from 5 interviews ($M = 105.2$, $SD = 12.6$). All sessions were recorded using a web application that facilitated collaborative annotation of noteworthy moments for later review. Initially, the interview recordings were reviewed, with key insights highlighted. Key turning points and relevant segments were identified using timestamps, and these insights were subsequently revisited and examined by the research team. Excluding the interviewer's introductory remarks and off-topic discussions, the corresponding audio recordings were transcribed.

The analysis process began with data familiarization through transcript review, followed by systematic initial coding where meaningful segments of text were identified and labeled. The third phase involved generating initial themes by examining relationships between codes and clustering related concepts. These candidate themes were subsequently reviewed against both the coded extracts and the complete dataset to ensure coherence and representativeness. In the fifth phase, themes were refined, clearly defined, and named to capture their core essence. The final phase involved producing the analytical report with illustrative data extracts. Complete documentation of the analysis process, including open coding summaries, a comprehensive codebook, and the evolution and development of themes, is available as Supplementary materials.

5.1. Thematic structure

The following subsections present the five main themes that emerged from the analysis of how designers engaged with movement features when developing installation concepts while Fig. 7 visualizes the themes and subthemes interconnected structure. These themes reveal how *designer-interpretable* features influenced creative practice specifically

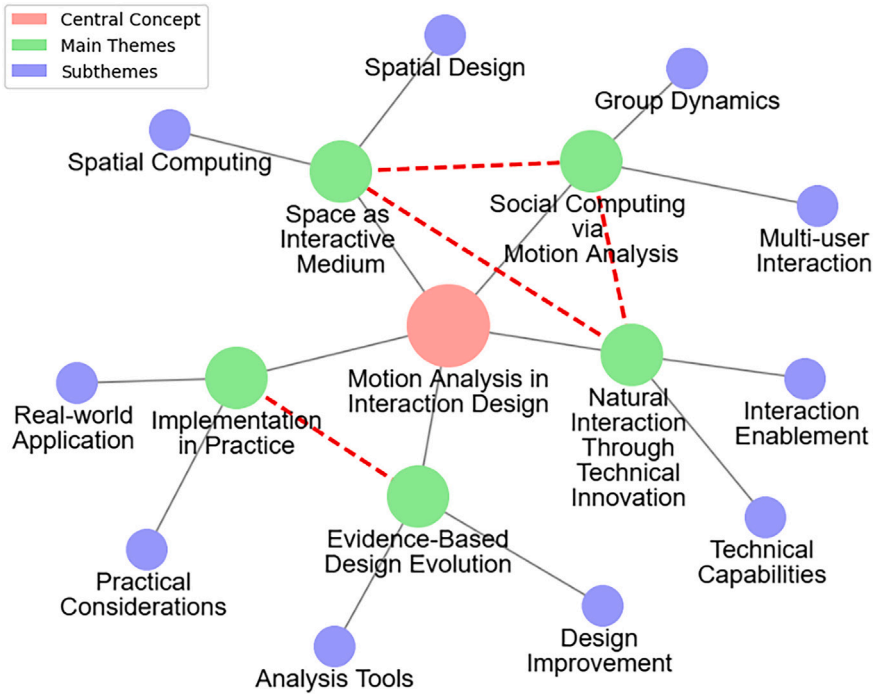


Fig. 7. Themes and subthemes in a interconnected structure. Dotted line represent main themes' relationships.

for installation design contexts. Companion code for theme development and evolution, thematic relationship analysis, and topic map visualizations is available as Supplementary materials.

5.1.1. Natural interaction through technical innovation

Through our analysis, we identified a significant pattern in how technical capabilities enable and shape natural interaction possibilities. This theme emerged primarily through the consistent attention participants gave to the relationship between technical sophistication and interaction naturality. Participant response patterns revealed distinct preferences across movement features: trajectory and heading were considered fundamental by three participants (P1, P3, P5), with P1 stating: *"I feel that trajectory and heading are fundamental for me, like having the basics,"* while heading alone was valued for predictive capabilities by four participants (P1, P2, P4, P5), as P5 emphasized: *"being able to know where the person is going allows me to anticipate many things."* Movement-feature specific applications demonstrated sophisticated technical understanding: P1 applied multi-joint tracking for complex gestural analysis: *"I could decide to take the trajectory of various joints from the same user, for example, head and both hands,"* while P5 utilized heading for temporal prediction requiring: *"a little work, making a prediction a few moments into the future."* Concrete interaction examples included P1's development of prototype-based behavior analysis where users interact naturally with cardboard environments while the system records poses for later implementation: *"you put users and nothing is happening in the system, you just have the scenario there, in cardboard, and from how users are reacting with the system, you start extracting the poses that will later define the final system."*

The theme encompasses significant challenges and limitations that constrain technical innovation. Scalability emerged as a critical barrier, with P5 identifying that *"as the number of people increases, the complexity doesn't grow linearly, but rather becomes almost exponential."* Technical precision versus practical implementation created persistent tension, exemplified by P4's comparison: *"motion capture has benefits over camera tracking—it's more precise, doesn't depend on lighting, but setting up those suits is horrible, for each person entering, 20 minutes not counting."* System calibration difficulties were highlighted by P1's observation about clustering sensitivity: *"I suppose one could calibrate the clustering sensitivity, because sometimes it seems like it's just one group, but in reality they're quite separated."* Novel movement-specific insights revealed fundamental differences from traditional HCI approaches: motion-based natural interaction requires temporal intention commitment, where users need time to commit to movements before triggering interactions, spatial tolerance parameters that balance accuracy with user comfort, and gestural ambiguity resolution that accommodates natural movement variation. Unlike traditional interfaces, movement systems enable unconscious interaction triggers and require higher reliability thresholds because system failures appear as environmental failures rather than user errors, fundamentally altering the relationship between technical capabilities and interaction naturality.

5.1.2. Space as interactive medium

The transformation of physical space into an active interface element emerged as a distinct theme through our analysis of participants' responses to the computational prototype demonstrations. Participant response patterns demonstrated universal spatial reconceptualization, with all five participants immediately conceptualizing space as reactive to user presence and movement, while four participants (P1, P2, P3, P5) specifically connected movement directions to conceptual navigation. Movement-feature specific applications revealed sophisticated spatial thinking: P1 applied heading detection for temporal navigation (*"if you move forward you go to the future and if you walk backward you go to the past"*), P2 utilized bounding box variations for environmental control (*"imagine there are projections on the walls, and you can control the water level depending on whether you make yourself taller or smaller"*), and P4 employed clustering for zone-based environmental adaptation where *"depending on how these groups organize in these zones, I can change the environment or put directed music with a speaker."* Concrete interaction examples included P5's cosmic environment concept where *"arm movements control galaxy gravity"* and *"those large arm movements end up being very fluid and can create complex interactions,"* and P2's adaptive museum system that redistributes engaging content across multiple displays based on crowd formation patterns to *"facilitate traffic."*

Challenges and limitations revealed complex spatial interaction management requirements. Multi-user coordination complexity emerged as

a significant barrier, with P1 noting: *"the system has to guide them a lot—how to tell them to separate or group?"* Environmental adaptation challenges included managing varying lighting conditions and user behaviors that affect tracking reliability, while spatial calibration complexity required sophisticated multi-camera system setup for accurate positioning. Scale management difficulties were highlighted by P4's observation about *"many people in an interactive experience"* requiring algorithms that handle exponential complexity growth. Novel movement-specific insights distinguished spatial computing through motion analysis from traditional spatial interfaces: movement enables vertical stratification of interactive space, allowing users to simultaneously engage with distinct planes within the same physical environment, creating layered interaction possibilities unavailable in traditional spatial computing. Physical movement becomes directly coupled to conceptual space navigation, enabling intuitive metaphorical interactions that bypass traditional interface paradigms. Groups of users can collectively reshape spatial properties through coordinated movement patterns, while spaces can respond to movement intentions before explicit actions occur, creating anticipatory environmental behaviors. This ambient spatial responsiveness represents a fundamental advancement where space becomes a predictive medium rather than a reactive surface.

5.1.3. Social computing via motion analysis

This theme emerged through participants' consistent attention to collective behavior and group interaction possibilities enabled by motion analysis. Participant response patterns revealed universal recognition of social computing potential, with all five participants acknowledging clustering capabilities for social applications, while three participants (P1, P2, P5) specifically designed competitive group experiences and four participants (P1, P2, P3, P4) discussed collaborative interaction enhancement. Movement-feature specific applications demonstrated sophisticated social behavior detection: P5 applied clustering as *"a base on which I can start developing multi-group experiences"* by extracting clusters and performing trajectory comparisons between groups, while P4 utilized parametric cluster control to define *"how people will have to interact with each other to generate a cluster"* through distance threshold manipulation. Concrete interaction examples included P2's adaptive content distribution system that replicates engaging content across multiple locations based on crowd formation (*"distribute people between screens to facilitate traffic"*), and P4's predictive social orchestration where the system *"can prepare something for when they're going to be there"* based on anticipated user convergence patterns.

Challenges and limitations revealed significant social computing complexity. Multi-user coordination difficulties were exemplified by P1's concern: *"the system has to guide them a lot—how to tell them to separate or group?"* while scale-dependent effectiveness was highlighted by P1's observation that success *"depends a lot on the target audience, this type of clustering can work quite well, but depends much on the designed experience."* Social behavior unpredictability created design challenges, as P2 noted the fundamental persistence of social dynamics *"whether people know each other or not,"* requiring systems to accommodate varying engagement patterns and personality types. Real-time group management complexity emerged through P4's recognition of difficulties in tracking multiple users simultaneously for coordinated experiences. Novel movement-specific insights revealed unique characteristics of motion-based social computing: implicit social engineering becomes possible through movement pattern encouragement without explicit instruction, enabling organic group dynamics that feel natural rather than imposed. Behavioral synchronization detection allows systems to identify and respond to unconscious movement coordination between strangers, creating enhanced social experiences. Social space optimization through motion analysis enables real-time redistribution of social attention to prevent overcrowding while maintaining engagement. Technical detection of group formation enables amplification of natural social behaviors through environmental feedback, representing a paradigm shift from designing explicit social interactions to augmenting and responding to natural social patterns in ways that feel both enhanced and authentic.

5.1.4. Evidence-based design evolution

Our analysis revealed a consistent pattern in how motion analysis provides designers with objective data to inform their design decisions. Participant response patterns demonstrated universal recognition of analytical value, with all five participants emphasizing objective behavioral measurement over self-reported feedback, while four participants (P1, P2, P4, P5) specifically discussed pattern detection capabilities and three participants (P2, P3, P4) explicitly connected analytical insights to design improvement processes. Movement-feature specific applications revealed sophisticated analytical thinking: P1 applied trajectory analysis for usage pattern identification and prototype-based research where *"you have the data, as a whole—you have ten tests and ten users touched their heads, so what happens if touching your head becomes the main interaction?"* P4 utilized multi-feature analysis for engagement measurement, stating: *"this allows me in a very objective way to see where they really spent more time,"* while P5 employed clustering analysis for understanding group formation patterns over time. Concrete interaction examples included P1's adaptive hotspot generation system that dynamically moves attraction points based on historical usage patterns, and P4's behavior-responsive environments that adapt based on detected engagement levels rather than user reports.

Challenges and limitations revealed significant analytical complexity. Data interpretation difficulties increased with scale, as P4 noted: *"right now with the videos it's super simple to see who has similarities because there are only three, but in an interactive experience there could be many people."* Historical data requirements created temporal constraints, with P1 recognizing that some analytical approaches require baseline data collection before meaningful insights can be generated. Real-time processing limitations challenged system capabilities, while analysis scalability was constrained by exponential complexity growth with increasing user numbers. Novel movement-specific insights distinguished evidence-based design through motion analysis from traditional evaluation methods: behavioral truth versus reported experience emerges as motion data reveals authentic user behavior that often contradicts self-reported experiences, providing more reliable design insights. Temporal engagement patterns become visible through movement analysis, capturing micro-patterns of interaction that are invisible to traditional evaluation methods and revealing engagement evolution over time. Spatial attention mapping through physical movement provides granular attention data that enables precise spatial interaction optimization unavailable through other means. Predictive design iteration becomes possible as historical movement patterns enable proactive design adjustments before user frustration occurs, representing a fundamental advancement in design methodology that enables optimization based on behavioral prediction rather than reactive problem-solving based on reported issues.

5.1.5. Implementation in practice

This theme emerged through participants' consistent attention to practical deployment considerations, revealing how designers balance theoretical possibilities with implementation realities when considering motion-based interaction design. Participant response patterns revealed universal implementation awareness, with all five participants discussing setup simplicity requirements and four participants (P1, P2, P4, P5) emphasizing cost-effectiveness as a determining factor, exemplified by P5's statement: *"the fact that it's so affordable, because it doesn't imply a super high entry cost."* Movement-feature specific applications demonstrated practical preference patterns: P1 emphasized bounding box and clustering simplicity (*"those seem very simple to me, but appear supremely powerful within that simplicity"*), P4 compared tracking technologies for context-appropriate selection, and P5 prioritized camera-based systems for high-throughput installations while recognizing motion capture for controlled environments. Concrete interaction

examples included P1's museum-ready installations prioritizing camera-based tracking for minimal user setup requirements, P4's hybrid tracking systems combining different technologies based on installation context, and P5's cost-effective educational installations using simplified feature sets for maximum deployment feasibility.

Challenges and limitations revealed implementation-specific barriers that distinguish movement systems from traditional interfaces. Setup complexity emerged as a critical constraint, with P4 highlighting motion capture difficulties: *"setting up those suits is horrible, for each person entering, 20 minutes not counting,"* while technical reliability requirements exceeded traditional interface standards because movement system failures appear as environmental failures rather than user errors. Infrastructure dependencies created additional constraints, as P4 noted potential *"problems with the signal"* in WiFi-dependent systems, while context adaptation challenges required technology selection based on deployment environment and target audience. Scale-dependent viability was constrained by exponential complexity growth with increasing user numbers. Novel movement-specific insights revealed unique implementation characteristics: technology-context coupling becomes fundamental as movement tracking technology choice becomes inseparably linked to deployment context in ways traditional interfaces do not experience. User setup invisibility emerges as a critical requirement, where properly implemented movement systems must require zero user preparation, unlike traditional interactive systems that depend on user setup procedures. Reliability threshold criticality exceeds traditional interfaces because movement failures appear as system rather than user failures, requiring higher operational standards. Scalability complexity demonstrates non-linear growth, necessitating fundamentally different system architecture approaches compared to traditional interfaces that scale more predictably. These implementation requirements represent novel constraints that fundamentally shape which technological approaches remain viable across different deployment contexts, creating implementation considerations that are unique to movement-based interaction systems.

5.2. Overview of interview findings

A cross-participant thematic synthesis revealed patterns of agreement regarding motion analysis system characteristics and implementation considerations. As illustrated in Fig. 8, universal agreement emerged across several areas, with all participants mentioning cost-effectiveness, user setup simplicity, and natural interaction characteristics when evaluating the system. Participants consistently referenced economic accessibility and minimal user preparation requirements when discussing practical deployment feasibility.

All participants recognized natural interaction capabilities, specifically noting systems that operate without requiring explicit user awareness or conscious interaction effort. Participants universally conceptualized spatial interaction applications and social computing possibilities when presented with motion analysis capabilities, indicating recognition of these domains as relevant application areas for movement-based systems.

Four out of five participants expressed a preference for objective behavioral data over subjective feedback methods when evaluating visitor experiences. As shown in Fig. 8, four participants noted potential advantages of motion analysis compared to traditional interview-based evaluation approaches. Fig. 9 presents feature preferences, revealing hierarchical patterns with heading detection, bounding box, and clustering each being mentioned favorably by four out of five participants. Participants universally discussed combining multiple features rather than using individual descriptors, with feature integration recognized by all participants as necessary for comprehensive visitor movement analysis.

To provide a comprehensive view of feature usage patterns, Table 2 connects each movement-based feature to its design purpose, actual participant usage during the study, and the primary themes where each feature appeared prominently. This mapping validates our framework's three dimensions while revealing hierarchical adoption patterns aligned with the framework's progressive complexity. Fundamental spatial features—trajectory, heading, and bounding box—were universally valued and appeared prominently in Natural Interaction and Space as Medium themes, validating their role as accessible entry points. Group detection features—clustering and hotspots—dominated Social Computing and Evidence-Based Design themes, with participants like P5 using clustering for formation analysis and P1 applying hotspots for adaptive design systems. Advanced coordination features—trajectory similarity and movement correlations—were recognized for social applications but less frequently applied, suggesting these require additional accessibility work despite sociological grounding.

Participants also identified implementation constraints alongside positive assessments. All participants acknowledged scalability challenges, noting that system complexity increases non-linearly with user numbers. Four out of five participants mentioned potential advantages for large group analysis compared to manual observation methods, while also recognizing system limitations. System calibration difficulties and environmental adaptation requirements were reported by three to four

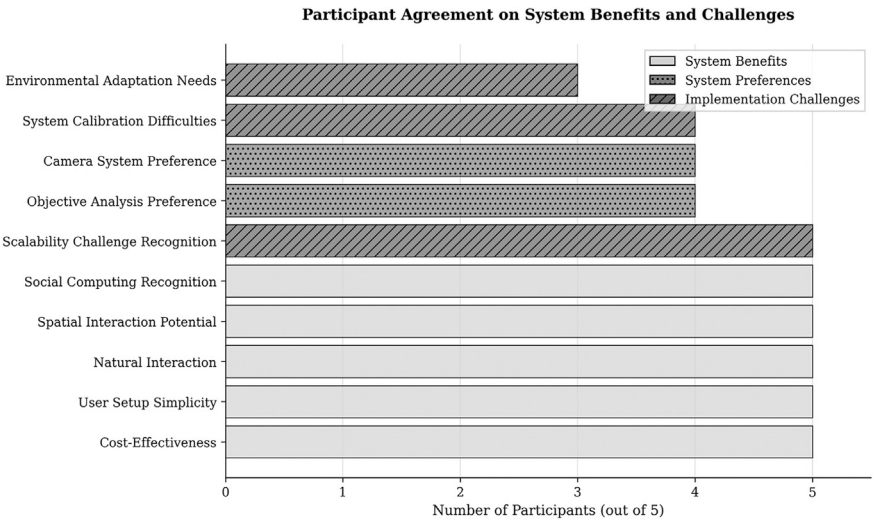


Fig. 8. Participants expressing agreement with key system benefits and recognizing implementation challenges.

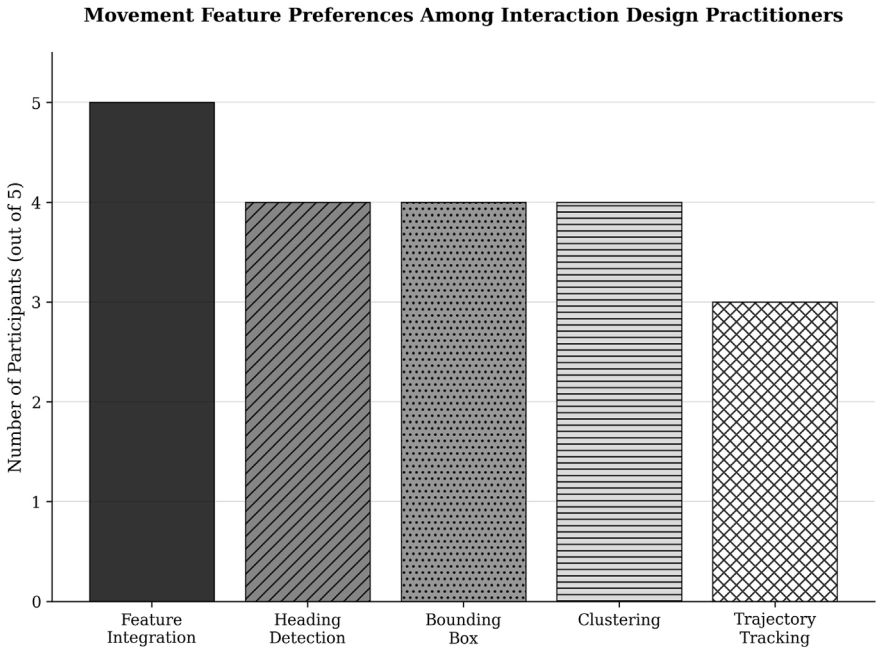


Fig. 9. Participant preferences for specific movement features based on explicit statements of value and utility.

Table 2
Movement feature usage patterns, design purposes, and thematic connections.

Feature	Framework purpose	Participant usage	Primary themes
Bounding Box	Analyze spatial occupancy; measure volume and physical presence	**P2, P3, P4, P5**: Fundamental spatial reference **All**: Combined with other features for comprehensive analysis	Space as Interactive Medium
Trajectory	Analyze movement in relation to space; detect usage patterns over time	**P1, P3, P5**: Fundamental spatial understanding **P1**: Usage pattern identification, adaptive systems **P4**: Engagement measurement	Evidence-Based Design Evolution; Space as Interactive Medium
Heading	Discriminate between actions; predict directional intent and future position	**P1, P2, P4, P5**: Predictive capabilities **P5**: Temporal prediction (“anticipate many things”) **P1**: Combined with trajectory for gestural analysis	Natural Interaction Through Technical Innovation
Instantaneous Clustering	Identify relationships among users; detect real-time group formation	**P5**: Group formation patterns over time **All**: Social interaction understanding **P4**: Combined with hotspots for engagement	Social Computing via Motion Analysis
Hotspots	Detect people flow; identify traffic patterns and frequently visited areas	**P1**: Adaptive hotspot generation system **P4**: Spatial engagement analysis **P1, P4**: Historical pattern analysis	Evidence-Based Design Evolution; Space as Interactive Medium
Trajectory Similarity	Analyze collective movement; detect coordinated behaviors across users	Recognized for social applications **P4**: Noted complexity with scale (“difficult with many people”)	Social Computing via Motion Analysis
Movement Correlations	Understand synchronized movements; identify unconscious coordination	Acknowledged value but less frequently applied Complexity noted for real-time design exploration	Social Computing via Motion Analysis

of the five participants, indicating awareness of practical deployment challenges that require consideration in implementation planning.

6. Discussion

6.1. Key themes in motion-based feature usage

6.1.1. Natural interaction through technical innovation

Our research reveals that interaction designers prioritize user agency and practical implementation when integrating movement-based features. This finding represents a significant distinction from existing research conducted in expert domains, where the focus has typically been on optimizing movement representations for individuals with extensive training in movement performance (Swaminathan et al., 2009; Subyen et al., 2011; Silang Maranan et al., 2014). Instead, our study addresses the broader field of interaction design, where end-users often possess varying degrees of familiarity with digital interfaces. This user-centered approach aligns with the contemporary understanding of HCI, where good interaction design maximizes simplicity and naturalness while minimizing cognitive load (Moere and Purchase, 2011; Dimara and Perin, 2019).

The emphasis on multimodal interaction emerged as a key aspect of natural interaction design. Our findings suggest that considering diverse input modalities, including body movement, head tracking, and gait recognition, creates more inclusive and adaptable interaction systems. This multimodal approach fills a significant gap in traditional interaction design, where single-modality interactions often fail to accommodate diverse user needs and preferences (Blair-Early and Zender, 2008). Participants consistently emphasized the importance of intuitive interaction, supporting previous research by Mentis and Johansson (2013) on the significance of natural movement in interaction design.

The preference for features that enable intuitive user actions, such as using a sweeping gaze to trigger an overview, demonstrates how designers seek to align technological capabilities with users’ habitual behaviors. This approach echoes recent work on direct manipulation interfaces, where the goal is to minimize the divergence between user intent and system response (Zhu et al., 2024). As noted by one participant,

"I would choose a feature that helps users understand what they need to do without too much instruction so that the interaction feels natural and intuitive." This perspective represents a shift from traditional expert-focused systems toward more inclusive, user-centered design approaches.

6.1.2. Space as interactive medium

The use of space as an interactive medium emerged as a key theme in our findings. The implementation of multi-view 3D pose estimation enables a comprehensive understanding of user movement within spatial contexts, supporting the concept of embodied interaction in HCI (Loke and Robertson, 2009; Schaper and Pares, 2016; Mishra and Cafaro, 2018). This aligns with recent work on situated visualization (Bressa et al., 2021), where interactions are embedded within real-world contexts to enhance user acceptance and engagement. The importance of contextual design was particularly evident in our findings, as interactions must be socially acceptable in public environments, considering both the user's perspective and that of observers (Montero et al., 2010).

Notably, our findings revealed that the prototype would be most beneficial in scenarios involving a large number of users, particularly in settings such as museums, events, or concerts where traditional technologies like motion capture may struggle to keep pace with the dynamics of large, unstructured audiences. This emphasis on spatial context reinforces the necessity of "situated interaction" (Zhu et al., 2024), where designs must be appropriate for their specific environment while maintaining flexibility to accommodate diverse user behaviors and environmental conditions (Blair-Early and Zender, 2008).

Our findings suggest that spatial awareness plays a vital role in creating effective interactive experiences. As one participant noted, *"It is an advantage to have cameras. I find the fact that large spaces can be covered very interesting."* This observation supports existing research on the importance of spatial configuration in interaction design, particularly in public environments where social acceptability must be carefully considered (Kules et al., 2004; Müller et al., 2012; Mishra and Cafaro, 2018).

6.1.3. Social computing via motion analysis

The study reveals significant implications for social computing through motion analysis, particularly in understanding group dynamics and multi-user interactions. The automatic identification of patterns and group behaviors through clustering features provides designers with powerful tools for rapid analysis of social interaction patterns. This capability is particularly valuable in large-scale implementations, where manual analysis would be impractical, similar to findings by Marshall et al. (2011) who studied how groups approach walk-up-and-use interfaces. This is evident in public spaces where multiple users interact simultaneously (Peltonen et al., 2008), and in scenarios requiring analysis of complex group dynamics like collaborative writing (Larsen-Ledet and Korsgaard, 2019) or multi-user smart home environments (Geeng and Roesner, 2019). Studies have shown that understanding group formation and interaction patterns becomes increasingly challenging as the number of simultaneous users grows (Russell et al., 2002), highlighting the importance of automated pattern recognition tools.

Our research bridges artistic exploration and analytical needs through feature visualization in movement-based interactions, taking a distinct analytical stance compared to conventional visualization methods that often fall short when capturing nuanced movement dynamics. Feature visualization proves to be a powerful tool for interaction designers, creating a feedback loop that fosters iterative and user-centered design practices, aligning with research on collaborative coupling in multi-user scenarios (Tang et al., 2006). The ability to visualize and analyze social interactions in real-time supports the development of more responsive and adaptive systems that can accommodate diverse user groups and interaction patterns (DiMicco et al., 2007). This approach enables designers to create flexible visualization systems where users can express complex interaction patterns and organize elements freely, supporting both individual and collective interaction needs, similar to findings on territoriality in collaborative workspaces (Scott et al., 2004).

Our findings indicate that visualization tools play a crucial role in understanding group formation and interaction patterns. As noted by one participant, *"Without visualization, it is very difficult for me to really know how interaction groups are formed."* This observation supports previous research by Yuill and Rogers (2012) on the mechanisms for collaboration and the importance of visual feedback in understanding collective behavior patterns. The significance of visual tools in understanding group dynamics is further emphasized in studies of collaborative spaces (Scott and Carpendale, 2010), where visualization helps clarify which areas are available for individual or joint task work. Research by Wang et al. (2017) demonstrates how visual representations of collaborative activities over time help users understand participation patterns and contributions. This aligns with findings from Niemantsverdriet et al. (2016), who showed that making group activities visually apparent through social translucence frameworks improved coordination in multi-user scenarios.

6.1.4. Evidence-based design evolution

The role of evidence-based design has emerged as a critical factor in the evolution of movement-based interactions. Our findings demonstrate how feature visualization enables designers to compare their initial concepts with actual user movements, supporting an iterative design process. This aligns with recent movement-based design approaches that emphasize the importance of iterative, experiential prototyping (Márquez Segura et al., 2016a). The accessibility and immediacy of our prototype remove barriers in the early stages of interaction design, fostering creativity and encouraging practitioners to explore unconventional ideas. This user-centered approach to design emphasizes the importance of understanding user intent, cognitive activities, and the broader context of interaction (Márquez Segura et al., 2016b; Wilde et al., 2017).

Furthermore, our findings indicate that the prototype's utility transcends disciplinary boundaries, as noted by one participant who stated, *"The prototype is very useful for me to ground the idea and take it to other fields of ideation."* This cross-disciplinary potential is enhanced by the system's ability to maintain flexibility while ensuring usability and practicality. This aligns with recent work on movement-based design methods that emphasize their value in multidisciplinary contexts involving both experts and novices (Vega-Cebrián et al., 2023). The feature visualization supports both detailed analysis and creative exploration, allowing designers to iterate on their ideas while maintaining a focus on user needs and environmental constraints. These features serve as experience prototypes or "quick and dirty" mock-ups that allow participants to get a sense of the target experience (Buchenau and Fulton Suri, 2000). This approach aligns with the contemporary understanding of interaction as an active, human-driven exploration where knowledge is constructed and refined through interactive manipulation (Márquez Segura, 2016). This approach resonates with recent work on embodied design that emphasizes the importance of combining physical engagement with play and playfulness to induce a creative mindset (Márquez Segura et al., 2016b).

The study reveals that designers value concrete decision-making tools that provide actionable insights. As one participant explained, *"By observing a trajectory, I can be very sure of where the user has passed, and I can make redesign decisions with certainty based on an analysis over time."* This finding supports existing research on the importance of data-driven decision-making in interaction design (Agrawala et al., 2011). The value of concrete tools also aligns with recent work on documenting and analyzing movement in design processes (Vega-Cebrián et al., 2023), where video recording and body maps are leveraged not only as evidence for evaluation but also as creative mediums for prototyping ideas. Furthermore, our findings echo a methodology that emphasizes the significance of the first-person perspective of movement in design (Loke and Robertson, 2013), suggesting that tools supporting direct

observation and analysis of movement can enrich the design process and lead to more informed decisions.

6.1.5. Implementation in practice

Our research highlights important practical considerations in deploying motion-based interaction systems. Designers consistently emphasized the need for balance between precision and practicality, often favoring consumer cameras over more complex sensing technologies. This preference reflects broader critiques that many movement interfaces fail not due to a lack of sophistication but due to poor interaction design that doesn't consider practical usability (Norman, 2010). Such emphasis on simpler technologies demonstrates a broader understanding that successful interaction design must prioritize features that are both functional and easy to replicate, with selected features chosen based on usability and practicality. Even when employing advanced approaches like machine learning for movement recognition, ensuring that models remain interpretable and understandable to designers should be prioritized over implementing more powerful but opaque algorithms, as the ability to effectively guide and understand the system's behavior ultimately leads to more successful interactions (Gillies, 2019).

A recurring theme in our discussions was the preference for implementation simplicity over the pursuit of extreme accuracy in interactive system development. As one participant emphasized, *"Obviously, its use comes with a sacrifice issue in the precision of the movement."* Previous work supports this observation, demonstrating that users think about movement in holistic terms rather than precise technical measurements (Gillies et al., 2015). The findings suggest that the emphasis should be on capturing broader patterns and dynamics of user movement rather than precise tracking. Studies of movement-based musical interfaces have found that users prefer interactive testing focused on usability over traditional technical metrics of accuracy (Fiebrink et al., 2011). This indicates that system reliability and interpretability often outweigh technical precision and sophistication in movement interaction systems. This finding ties directly to the concept of flexibility in visualization systems, where interactions should allow users to express complex data queries and organize elements freely while maintaining system reliability and ease of implementation (Elmqvist et al., 2011).

The study reveals that successful implementation often depends on striking the right balance between technical capability and practical usability. As noted by one participant, *"It was much easier and people preferred the abstract interaction of a bounding box, because it's less frustrating."* Another participant emphasized this principle from a design perspective: *"When designing experiences, the simpler the better. Because if we're talking about experiences where you have to interact with a system using your body… you have very little time for the person to understand how the system works."* These observations align with research in movement-based interfaces for dance and public installations, showing that the appropriate balance between technical sophistication and practical implementation varies depending on the specific context and goals of the system (Halpern et al., 2011; Fdili Alaoui et al., 2015). These findings indicate that successful movement interaction design requires careful consideration of practical implementation constraints from the earliest stages of design. Rather than pursuing technical sophistication for its own sake, designers should focus on creating reliable, interpretable systems that support users' natural understanding of movement while remaining practically implementable with available technology.

6.2. Implementation scenarios and implications for interaction design

Our multi-view 3D pose estimation approach reveals significant implications for deploying movement-based interactions in real-world contexts. The technology's capacity to handle large and unstructured groups through automated clustering algorithms addresses a fundamental limitation in current interaction design practice. As one participant noted, *"In massive use settings, where users are expected to pass quickly, cameras should be used. For example, in a museum, an event, or a concert."* This capability to automatically identify and classify movement patterns through clustering algorithms enables designers to understand group dynamics in real-time without the manual analysis that becomes impractical with multiple simultaneous users.

The ability to identify traffic patterns and interaction hotspots represents a paradigm shift in environmental design, building upon existing research on situated visualization (Bressa et al., 2021) while introducing new possibilities for responsive space design. Our hotspot detection and trajectory analysis features enable designers to optimize spatial layouts based on empirical data rather than assumptions, supporting the concept of evidence-based design evolution. This capability proves particularly valuable in environments such as museums and concert venues, where understanding natural congregation patterns and movement flows is crucial for experience design.

The integration of group behavior detection through clustering and trajectory similarity analysis facilitates cross-disciplinary collaboration by creating a common language between technical teams and designers. Our social computing features—including instantaneous clustering and correlations across movement patterns—bridge the communication gap that often hinders collaborative design processes. As one participant emphasized, *"Without visualization, it is very difficult for me to really know how interaction groups are formed,"* highlighting how these complex detection capabilities enable shared understanding of social dynamics regardless of technical background.

Beyond these immediate implementation benefits, our findings reveal fundamental shifts needed in how the HCI community approaches movement-based interaction design. The documented gap between technical capability and design accessibility suggests that current research priorities may be misaligned with practitioner needs. While the computer vision community continues advancing pose estimation accuracy and algorithmic sophistication, our results indicate that accessibility and interpretability represent more pressing barriers to real-world deployment.

The emphasis our participants placed on user agency and natural interaction aligns with Jacob et al.'s (2008) framework for reality-based interfaces, which advocates for leveraging users' existing physical and social skills. Our findings extend this principle to the design process itself—designers also need tools that build upon their existing spatial and creative thinking skills rather than requiring new technical competencies. Our findings with group detection features suggest that complex technical capabilities can potentially support designers' natural reasoning about space and social interaction when presented through accessible interfaces. This suggests that future movement interfaces should systematically identify and incorporate both user skills and designer skills into their interaction models.

Our visualization approach for group behaviors demonstrates that technical capabilities can be made accessible without sacrificing analytical power, building directly on Gillies et al.'s (2015) work showing how interpretable visualizations can bridge technical and experiential aspects of design. Our spatial exploration paradigm extends their insights by demonstrating that designers can develop a deep understanding of group dynamics and traffic patterns through experiential manipulation rather than algorithmic parameter adjustment. The ability to visualize clustering results, hotspot formations, and trajectory similarities enables designers to grasp complex social patterns intuitively, suggesting that many areas of HCI could benefit from similar accessibility-focused visualization approaches.

The contextual design considerations our participants emphasized when working with group detection features connect directly to Bressa et al.'s (2021) work on situated visualization, but extend it to the design process itself. Our findings suggest that movement-based interactions must be designed for specific contexts from the outset, with our traffic pattern analysis and hotspot detection providing the empirical foundation for context-appropriate design decisions. This implies developing evaluation tools that can assess both technical performance and social

impact across different contexts, moving beyond laboratory-based evaluation to realistic deployment scenarios where large, unstructured groups naturally occur.

The diversity of designer backgrounds we identified challenges the field's implicit assumption that technical training is a prerequisite for sophisticated interaction design. Our approach to making complex group detection algorithms accessible to non-technical practitioners suggests a need for new research methodologies that explicitly include diverse designers as equal collaborators rather than end-users of technical systems. The field must develop evaluation criteria that prioritize designer interpretability alongside technical performance metrics, fundamentally shifting how we measure success in movement-based interaction research.

6.3. A roadmap for future movement descriptors

6.3.1. Understanding differential feature adoption

Our findings revealed hierarchical adoption patterns: fundamental spatial features (trajectory, heading, bounding box) achieved immediate uptake, while advanced coordination features (trajectory similarity, movement correlations) were recognized as valuable but less frequently applied. This gap indicates that coordination features have not yet achieved the same level of interpretability rather than being less useful.

The cognitive difference is substantial. Simple spatial features map directly onto observable phenomena, requiring minimal interpretation. Coordination features, however, require designers to simultaneously consider relationships between multiple users evolving over time. As P4 noted about trajectory similarity with many people, the cognitive load exceeds intuitive understanding capacity.

The solution involves applying accessible presentation principles more rigorously. Coordination features require intermediate visualizations decomposing complex relationships into digestible components—timeline visualizations showing how coordination evolves for specific user pairs, spatial heat maps connecting abstract metrics to concrete locations, and graph-based representations with semantic labels like *"parallel movement"* or *"mirroring"* rather than correlation matrices. The lesson from our Evidence-Based Design Evolution theme is clear: provide discrete, labeled events (*"Users A and B are moving in synchrony"*) rather than continuous similarity scores.

6.3.2. Expanding to semantic and affective features

Participants naturally employed semantic language describing movement: users *"exploring"* spaces, groups *"congregating,"* individuals appearing *"hesitant"* or *"confident."* These descriptors bridge raw movement data and design-relevant concepts, suggesting a promising direction for future features.

Semantic features would operate at higher abstraction levels while remaining grounded in objective measurements. Rather than requiring designers to interpret velocity variations, semantic features would provide labeled classifications directly supporting design reasoning. For instance, *"exploration"* could integrate spatial coverage from trajectory data, velocity variations indicating examination, and directional changes—presenting simply as *"User is exploring the space"* rather than displaying underlying metrics. Similarly, *"hesitation"* would analyze approach-avoidance patterns combined with heading variations, *"confidence"* through trajectory consistency and direct paths, and *"engagement"* through proximity patterns, postural indicators, and sustained heading toward targets.

The key principle is avoiding specialized movement knowledge requirements while providing meaningful behavioral insight, distinguishing our approach from Laban Movement Analysis systems which are relevant primarily to trained performers. Our semantic features must make sense for general public installation audiences.

Affective features represent a related category, though inferring affect from movement presents technical and ethical challenges. Future work should focus on observable behavioral indicators rather than claiming direct emotional inference—*"group cohesion"* measuring consistent spatial relationships without attributing emotional bonds, *"spatial comfort"* identifying distance maintenance without attributing specific emotions. Visualization should employ natural language labels, simple visual indicators, and event-based notifications.

6.3.3. Extending spatial feature categories

Our current spatial features provide substantial coverage, yet findings revealed unaddressed dimensions that participants' design concepts suggest would enhance analytical and creative work. Vertical spatial relationships emerged as a notable gap. Our Space as Interactive Medium theme documented P2's height variation concepts and participants' interest in *"vertical stratification of interactive space,"* yet no features explicitly capture vertical positioning. Future features analyzing crouching, reaching, or jumping behaviors would particularly serve projection mapping and spatial audio installations. Height variation analysis could extend to group comparisons, identifying when users make themselves taller or smaller relative to others, potentially indicating social dynamics.

Spatial formation features would extend clustering with socially meaningful patterns. While instantaneous clustering identifies proximity, it does not capture geometric arrangements indicating interaction types. Drawing from Setti et al.'s (2015) F-formations work, features detecting conversational circles, presentation formations, and linear arrangements would reveal whether groups engage in conversation, collectively view content, or simply co-locate without interaction—distinctions crucial for designing appropriate environmental responses.

Boundary interaction features address another dimension participants implicitly referenced. While hotspots identify frequently visited areas, features detecting boundary crossing patterns, edge interaction behaviors, and threshold hesitation would enable designers to understand how users mentally partition interaction spaces, supporting P2's adaptive museum concept that *"facilitates traffic."*

Spatial attention features would bridge position-based analysis with intentionality indicators. While heading provides directional facing, features identifying focal point attention, attention shifting between targets, and joint attention when multiple users orient toward the same target would support the anticipatory interaction design our Natural Interaction Through Technical Innovation theme revealed participants valued.

6.3.4. Temporal dynamics, feature composition, and evaluation considerations

Our features emphasize instantaneous or short-term windows, yet participants' concepts suggest longer-term temporal dynamics would provide valuable insights. Rhythmic pattern features detecting periodic behaviors—users repeatedly visiting locations or groups cycling between phases—would extend P4's interest in *"where they really spent more time"* by revealing cyclical patterns. Features analyzing return behavior, dwell patterns, and engagement trajectories would support Evidence-Based Design Evolution.

A critical finding was that participants universally combined multiple features rather than using them in isolation. Future work should explore meaningful composition while maintaining interpretability as combinations increase complexity. Layered presentation strategies where base features provide a foundation and composed features add enrichment—with progressive disclosure allowing complexity control—would support compositional thinking. For instance, integrating clustering with trajectory similarity identifies not just who forms groups but also whether they exhibit coordinated movement.

As the community develops new features, evaluation frameworks assessing technical validity, designer interpretability, and design utility become crucial. Future work should investigate how technical expertise, domain specialization, and cultural context affect feature interpretability across diverse designer populations, ensuring accessibility beyond technically-oriented practitioners.

7. Limitations

One significant limitation of our study lies in the computational prototype's workflow, which separates the capture and 3D pose estimation as a pre-processing stage. This division ruled out the realization of an end-to-end real-time solution for evaluating in-situ user behavior during the complete design and implementation cycle of interaction experiences. As a result, our study primarily focused on the feature extraction and visualization stage, missing the opportunity to assess the tool's performance in a comprehensive, hands-on context. The absence of real-time pose estimation limited our ability to observe how practitioners would integrate the technology into the complete process of interaction design, implementation, and user experience assessment. While our study showcased the value of *designer-interpretable* features during concept ideation and early design stages, it is important to recognize that a full-cycle evaluation would provide a more realistic and holistic perspective on the computational solution's applicability in end-to-end scenarios. Further research should aim to bridge this gap, exploring the intricacies of real-time applications and their impact on user experience in a broader, more dynamic context.

Our feature set addresses designers employing data-driven approaches for installations serving general audiences, deliberately excluding practices emphasizing expressive movement qualities, first-person embodied experience, or performance expertise. This scoping reflects our understanding that movement-based design encompasses diverse practices requiring distinct analytical tools: soma-design practitioners need first-person experiential features, choreographers require expressive quality notation, ergonomic assessors demand precise joint tracking with environmental context, and biomechanical analysts need temporal kinematic details our spatial features do not provide. Our contribution demonstrates designer-interpretable principles for one design practice context, not universal movement analysis applicability.

A critical gap in our research concerns our approach's consideration of diverse body types and mobility capabilities. While our prototype demonstrates effective movement analysis for typically-abled users, we did not explicitly examine how the system might function for users with different mobility capabilities or non-normative body configurations. Our multi-view 3D pose estimation relies on detecting standard skeletal landmarks, which may not accurately represent users who use mobility devices, have limb differences, or move in ways that deviate from typical bipedal locomotion patterns. The movement-based features we developed—particularly those requiring full-body tracking like bounding boxes and trajectory analysis—may not capture meaningful movement information for users with limited mobility or alternative movement patterns. This represents a significant oversight, as movement-based installations in public spaces should be designed to accommodate diverse abilities and body types.

While our qualitative methodology enabled deep exploration of designer experiences and needs when working on installation design challenges, we acknowledge the absence of controlled quantitative comparisons with traditional movement analysis methods or alternative design approaches. Our study extracted quantitative insights from the qualitative data, including unanimous agreement among participants on the accessibility benefits of our approach for installation contexts and systematic analysis of feature usage patterns during design sessions. However, a rigorous comparative evaluation would require a different study design that directly measures objective metrics such as design completion time for installation scenarios, accuracy of visitor pattern interpretation, number of design iterations, and quality of resulting interaction concepts when comparing our *designer-interpretable* approach against computational frameworks that prioritize technical sophistication over interpretability, or against basic data visualization tools. Such controlled comparisons would provide the objective evidence needed to quantify the advantages and limitations of our approach for installation design, making the findings more generalizable across different designer populations working in similar contexts and establishing clearer boundaries for when our approach provides advantages over alternatives.

8. Conclusions

This research makes three contributions to movement-based interaction design through the development and evaluation of *designer-interpretable* features for multi-view motion analysis systems supporting installation design for general audiences. First, we provide proof of concept evidence that *designer-interpretable* features can translate technical movement analysis into concepts accessible to non-technical practitioners working on installations serving diverse visitor populations. Seven features serve as examples demonstrating that group detection algorithms, traffic pattern analysis, and interaction hotspot identification can be made accessible without requiring technical expertise in computer vision or machine learning. These features specifically address visitor behavior analysis needs in installation contexts—understanding spatial navigation, social group formation, and engagement patterns—distinguishing our approach from systems designed for biomechanical precision, expressive movement qualities, or specialized performance assessment.

Second, we demonstrated that spatial visualization approaches enable designers to explore complex visitor movement patterns without requiring algorithmic expertise. Our prototype's three-component architecture—3D pose estimation, feature extraction, and visualization engine—provides evidence that technical data can be made accessible through designer-friendly interfaces optimized for installation design workflows. Our findings suggest that spatial exploration approaches may be more effective than parameter manipulation for enabling non-technical practitioners to understand group dynamics, social patterns, and traffic flows through intuitive visual interaction. This contribution is specifically relevant to designers employing data-driven, analytical approaches to installation design, complementing alternative methodologies like bodystorming, somaesthetic exploration, or choreographic methods that emphasize experiential and performative dimensions.

Third, we provided empirical evidence that this approach bridges the gap between technical capability and design practice for installation contexts. Our qualitative study with interaction designers revealed five themes demonstrating how *designer-interpretable* features facilitate the integration of visitor movement analysis into creative practice when designing for diverse audiences in room-scale environments. Participants used group detection and pattern analysis capabilities to inform their design decisions for installation scenarios, validating our theoretical framework's progression from technical reliability to creative exploration opportunities. The concrete design concepts participants developed—adaptive spatial experiences, social coordination installations, and evidence-based visitor guidance systems—demonstrate the specific types of interactions our approach supports while naturally illustrating the boundaries of our contribution.

These contributions collectively demonstrate that the accessibility gap between advanced computer vision capabilities and interaction design practice can be addressed through systematic attention to designer interpretability, providing a foundation for more inclusive and effective movement-based interaction design tools specifically for installation contexts serving general audiences. We acknowledge that our approach serves specific design practices and contexts rather than claiming universal applicability across all movement-based interaction design domains, recognizing that different practices—from ergonomic assessment to expressive performance—require different analytical approaches and feature sets tailored to their particular needs.

9. Future work

Several directions for future research emerge from this study, each building upon our current findings while exploring new possibilities for advancement in the field. Most critically, developing an end-to-end system that performs 3D pose estimation and feature extraction in real-time would address the current prototype's limitation of requiring pre-processing. Such an advancement would enable researchers to observe how interaction designers integrate motion-based features

during actual design sessions and collaborative workflows, providing more ecologically valid insights into professional practice. Real-time processing would also allow for studying how designers iteratively refine their understanding of movement through immediate visual feedback, potentially revealing new patterns in creative exploration.

Future research should investigate how pose estimation algorithms can be adapted to recognize assistive devices and alternative movement patterns, how movement features can be redesigned to be meaningful across different mobility capabilities, and how visualization approaches can represent diverse forms of human movement without privileging normative body configurations. This work is essential for ensuring that sophisticated movement analysis technologies contribute to inclusive rather than exclusionary interaction design practices.

Our approach to visualizing visitor behavior descriptors shows promise for extension to related domains beyond the installation contexts we studied. While we focused on museums, galleries, and interactive exhibitions, the principles of *designer-interpretable* features might transfer to other environments where understanding movement informs design decisions—retail spaces for optimizing layouts, educational environments for understanding collaborative behaviors, or healthcare facilities for improving wayfinding. However, such extensions would require careful investigation of whether our feature set addresses the specific analytical needs of these domains or whether alternative features would be more appropriate. Domain-specific research should examine how our framework's three dimensions might guide feature development for contexts beyond public installations.

Exploring how *designer-interpretable* principles might be adapted for alternative movement-based design practices represents another important direction. Our work addressed designers employing data-driven, analytical approaches, but future research could investigate whether similar accessibility principles might benefit practitioners using bodystorming, somaesthetic exploration, or choreographic methods. For instance, real-time feature visualization during bodystorming sessions might help designers compare their enacted movements to observed visitor patterns, while somaesthetic practitioners might benefit from features capturing first-person movement experiences differently than our third-person visitor analysis. Such investigations would clarify whether designer interpretability principles transfer across design practices or require fundamentally different tool support.

Additionally, future work should focus on developing more sophisticated feature extraction techniques and their visualization possibilities. This could include creating new algorithms for movement analysis, exploring machine learning approaches for pattern recognition, and developing more advanced visualization tools capable of handling increasingly complex movement data. Such advancements would enable designers to work with more nuanced aspects of movement and interaction, potentially uncovering new patterns and relationships that could inform more innovative design solutions. The integration of machine learning techniques, in particular, could help automate the identification of meaningful patterns and relationships in movement data, making the design process more efficient and effective.

CRediT authorship contribution statement

Antonio Escamilla: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Javier Melenchón:** Writing – review & editing, Methodology, Conceptualization. **Carlos Monzo:** Writing – review & editing, Methodology, Conceptualization. **Jose Antonio Morán:** Supervision, Methodology, Conceptualization. **Juan Pablo Carrascal:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijhcs.2026.103802 or at Mendeley Data (http://dx.doi.org/10.17632/5cvkx3gm9b.4). The prototype implementation, including source code and sample multi-view scene data with camera calibration information, is available at https://github.com/AntonioEscamilla/Exploring_movement_features_for_interaction_design. The repository enables immediate replication of the movement analysis capabilities and includes ready-to-use data for exploring movement-based features without requiring specialized capture equipment.

Data availability

Data will be made available upon request.

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